The Effects of Firm Closures on Worker Outcomes: Evidence from Pre-War Utah*

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Abstract

What was the role of firms in shaping workers' careers historically and how much did it matter when their firm closed? To answer these questions, we construct an annual linked employer-employee panel for 1900–1940 from the city directories of Salt Lake City and Ogden, Utah, which listed every adult resident with his occupation, employer and address. Transcribing 76 editions and 3.6 million listings, we harmonize employers across editions, link men across editions and, through FamilySearch, to the complete-count censuses, and then validate the panel against the census. We show that the firms played a major role in the pre-war urban economy. Most working-aged men were employed at a firm, and most career advancement occurred within firm. Using a stacked event study in which we compare men at failed firms to matched comparison men at surviving firms, we find that firm closure led to persistent negative effects on worker's labor market outcomes. Displaced men were 30 percentage points less likely to be listed with an employer a year later and 11 points less likely eight years later. One tenth moved away from the city and the rest were less likely to have steady employment and more likely to hold lower-status occupations. Marriage, fertility, and mortality were unaffected. Closures in 1930–1933, the early years of the Great Depression, were even more detrimental to worker outcomes.

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1 Introduction

Firms occupy a central place in the modern labor market. Matched employer-employee data show that where a worker is employed accounts for a large share of the dispersion in pay (Abowd, Kramarz and Margolis, 1999; Card, Heining and Kline, 2013; Song et al., 2019), and that losing a job when a firm closes costs workers a significant share of their earnings, with the heaviest losses falling on long-tenured workers and on those displaced during recessions (Davis and von Wachter, 2011; Jacobson, LaLonde and Sullivan, 1993; Lachowska, Mas and Woodbury, 2020). Whether firms mattered as much to workers a century ago is an open question. The early twentieth century was the age of the managerial corporation (Chandler, 1977) and of the first personnel departments and internal promotion ladders (Jacoby, 1985; Sundstrom, 1988). It was also an era of short jobs and high turnover, of few benefits and no unemployment insurance, and of an enduring ideal of the independent proprietor (Blackford, 2003; Brandes, 1976; Jacoby and Sharma, 1992; Mills, 1951; Owen, 1995). Historians have reconstructed careers inside particular companies from their personnel records (Carter and Savoca, 1990; Sundstrom, 1988), but data availability has limited researchers' ability to study historic employment dynamics at a large scale and at a high frequency. Historical research on careers has linked individuals across decennial censuses (Abramitzky et al., 2021; Ward, 2023), which observe individuals only once every ten years and record their occupation but not employer.

This paper constructs an annual linked employer-employee panel for the first four decades of the twentieth century by transcribing and linking city directories for two major Utah cities: Ogden and Salt Lake City. Each edition listed the city's heads of household with their name, occupation, employer, and home address. A second section listed every firm with their address grouped by industry (Albers and Kappner, 2023; Bühler, Cantoni and Weigand, 2024). We transcribe the 76 surviving editions of the Salt Lake City and Ogden directories for 1900–1940, covering 3.6 million listings, using a large language model.[1] We then clean and standardize occupations and firm names, geocode addresses, and then link firms and individuals across editions. Finally, we match individuals to FamilySearch profiles which we use to link them to the complete-count censuses of 1900–1940 (Buckles et al., 2023; Price et al., 2021).

Using this panel, we first describe the importance of firms in shaping careers in the pre-war urban labor market. Firms employed about 60 percent of working-age men, with the largest 2 percent of firms employing 40 percent of workers. Young men switched occupations and employers frequently, advancing towards higher paying occupations until around age

[1] We use Gemini Flash

40, before stabilizing. Most occupational advancement occurred within firms, although a significant fraction of employer moves were accompanied by an upward occupational switch. As in modern times, firms also varied in the wages and non-wage amenities they offered. Employer fixed effects explain an additional 8 percent of the variation in annual wages after controlling for job tenure and occupation, and a revealed-preference ranking of firms from workers' voluntary moves (Sorkin, 2018) suggests a substantial role for non-wage amenities as this firm value measure correlates only 0.42 with the firm's wage premium.

Next we study how firm closures affected workers on a variety of labor market, migration, and household outcomes. On average, 17 percent of firms closed each year, with increases during the Panic of 1907 and the early Depression. Using a stacked event study, we compare workers at failed firms to comparison men at surviving firms matched on age, tenure, occupation, firm size, and his career trajectory over the five years leading up to the firm closure. Within this matched sample, the two groups have parallel pre-trends on all of the outcomes we study.

We find that men whose employer closed were 30 percentage points less likely than matched comparison men to be listed with an employer in the following edition, with this gap declining to 11 points eight years later. A tenth of displaced men moved away from the city with three percent leaving the county. Those who stayed were divided between unemployment, weak attachment to the labor force and lower-status work, with the employed having occupational status 3 to 4 percent below their comparison men's, a gap that persisted over eight years. Using genealogical linkages we find no effects of firm closures on the probability of death, marriage, or on total fertility. Closures in the Depression years 1930–1933 were more detrimental to worker outcomes in the short run, with a 43 percentage point loss in employment and a 10 point loss in status in the first year compared with 28 points and 2 percent for closures in other years, although eight years later the two groups had converged.

The paper contributes to three literatures. The first studies firms in the historical labor market. From the records of individual companies, historians have established that long jobs, internal promotion and personnel policies existed in some large firms before the First World War and spread after it (Carter and Savoca, 1990; Jacoby, 1985; Jacoby and Sharma, 1992; Owen, 1995; Sundstrom, 1988). Our data show these features from the worker's side and across a whole labor market, including the small firms that company records never cover, and they add a measure of how much workers valued one firm over another.

Our second contribution is to the literature on displaced workers, which has documented large and persistent earnings losses in administrative data from the 1970s onward (Davis and von Wachter, 2011; Jacobson, LaLonde and Sullivan, 1993; Lachowska, Mas and

Woodbury, 2020; Sullivan and von Wachter, 2009). We extend it to an era without unemployment insurance for most of our period and to the Great Depression, finding similarly detrimental effects in this era.

Our third contribution is to the growing use of historical records beyond the census (Abramitzky et al., 2021; Albers and Kappner, 2023; Bühler, Cantoni and Weigand, 2024). City directories survive for hundreds of American cities and most years from the 1880s to the 1960s. We demonstrate their value and present a template which we hope can be used to transcribe and make available to research the entire corpus of U.S. city directories.

The rest of the paper proceeds as follows. Section 2 describes the construction and validation of the panel. Section 3 characterizes firms and careers in the two cities. Section 4 sets out the closure design and Section 5 the results. Section 6 concludes.

2 Construction of Employee-Employer Panel Data

We construct an annual panel of occupations and employers from the city directories of two major Utah cities, Salt Lake City (SLC) and Ogden, covering the years 1900–1940. City directories were published for many American cities from the late eighteenth century onward, serving as commercial reference works that helped merchants, salesmen, creditors, and the post office locate residents and firms. Publication frequency varied across cities and across time, but by the 1890s most larger cities received a new edition every year (Albers and Kappner, 2023). Each volume contained an alphabetical list of the city's adult residents with their full name, occupation, employer and home address. Listings focused on heads of household and skewed male (Albers and Kappner, 2023). A second section, the classified business directory, listed every firm under its line of trade with its address and, for incorporated firms, its officers. Figure 1 reproduces a page of each section from the 1920 Salt Lake City volume. R. L. Polk & Co. of Detroit became the dominant publisher of these volumes and continued to issue them well into the twentieth century. Most directories survive only as bound volumes or microfilm in local libraries; the Utah volumes we use, however, have been scanned by the University of Utah's Marriott Library and are available online.

Using a large language model, we transcribed page scans of each available SLC and Ogden Polk directory[2] from 1900–1940, 76 directories in total covering 3.6 million listings. For each listing we classified its printed occupation using the 1950 Census occupational classification, and geocoded its home address using historical street maps and Census Bureau

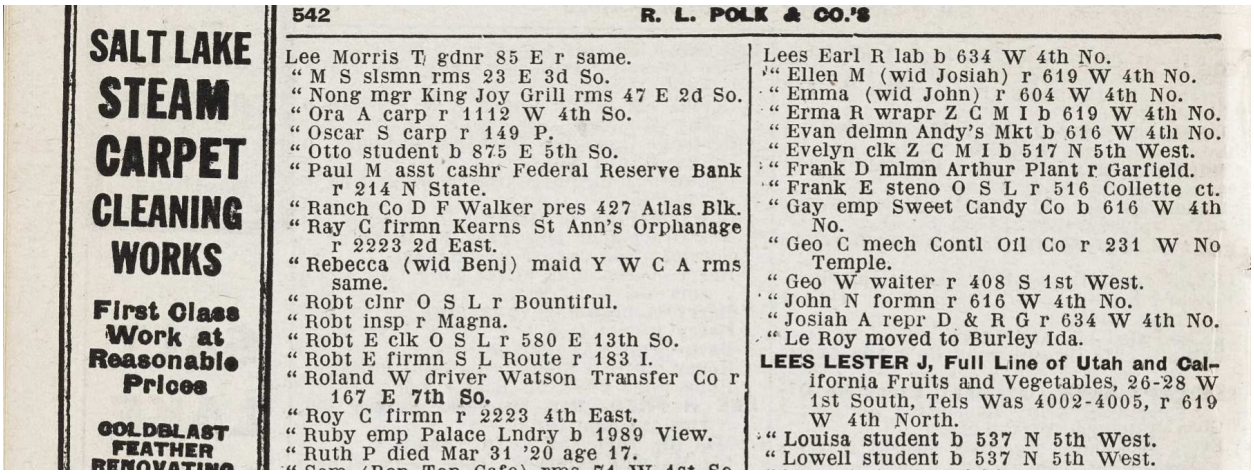
[2]Salt Lake City has a directory for each year in this window while Ogden is missing directories for six years: 1901, 1902, 1904, 1905, 1909 and 1937.

address ranges. Employer names, which Polk printed for more than half of the listings with an occupation, were harmonized across editions within each city. We then merged individuals across editions within a city on name and address, allowing an address change when the occupation or employer was unchanged or a family member moved with them, and linked each individual on name and location to a FamilySearch Family Tree profile. FamilySearch profiles include links to the complete-count censuses of 1900–1940 constructed by the Census Tree project of Brigham Young University's Record Linking Lab (Buckles et al., 2023; Price et al., 2021), which give us each person's birth year, birthplace and census occupation, and for 1940 their wage and weeks worked.

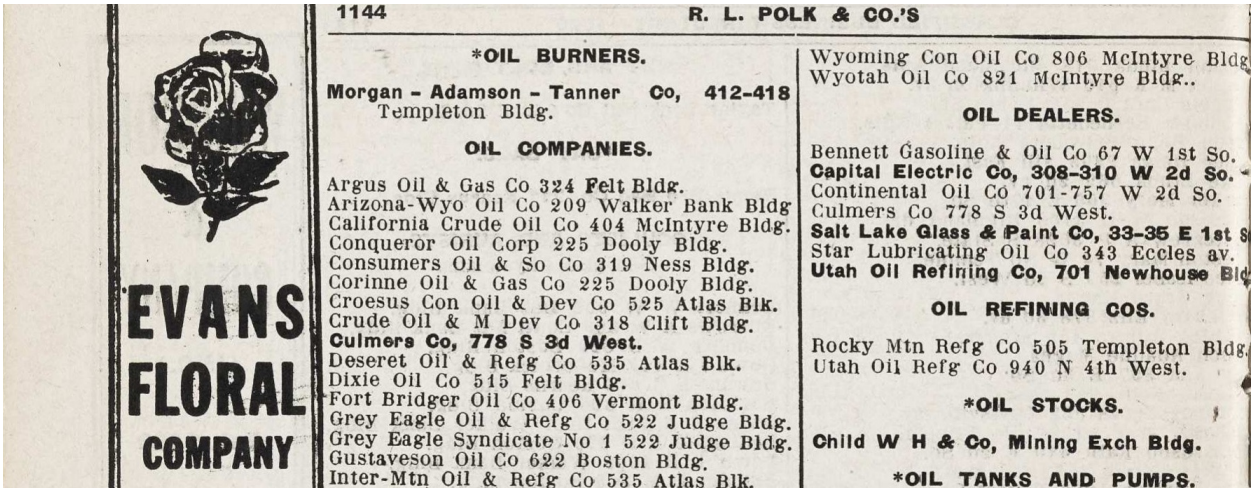
Table 1 provides summary statistics and linking rates for the two series, with every rate expressed as a share of the edition's listings averaged over the city's editions. The Salt Lake City book holds 70,600 listings in an average edition over its 41 editions and the Ogden book 19,500 over its 35. Three quarters of listings print an occupation, nearly all of which receive a census occupation code, and 42 to 43 percent print an employer, more than half of those that have an occupation. Approximately 95 percent of listings are accompanied by an address, with 70 percent of listings placed at a house-level point, 3 to 6 percent placed at a named building or on a street, and 4 to 11 percent in a suburban town. Between 10 and 16 percent of listings carry an address that is not successfully geolocated, often because the address belongs to an outlying settlement or is given relative to a cross street. Before an address, directories print a modifier indicating a person's occupancy status. Approximately 30 percent of listings are marked as roomers or boarders and 63 percent as residents or householders; because the books of 1930 and later no longer mark roomers and boarders, their share before 1930 is 42 percent. Between 62 and 64 percent of listings are linked to a FamilySearch profile, and through the profile between 38 and 52 percent to a given census; the census rows fall away from the FamilySearch row toward the ends of the period because not everyone listed in 1900 was alive in 1940 and not everyone listed in 1940 was born in 1900.

2.1 Occupations

Ogden and SLC directories print occupations for approximately three quarters of listings (74 and 73 percent). These were harmonized to the 1950 Census occupational classification (OCC1950) using mappings implied by individuals who were observed in both the census and a city directory in the same year. For such individuals, we observe both the printed string and the census occupation. Because occupation strings may refer to different OCC1950 codes depending on the employer's industry and because meanings of occupations change over time,



(a) Alphabetical list of residents (top of p. 542)



(b) Classified business directory (top of p. 1144)

Figure 1: Two pages of *R. L. Polk & Co.'s Salt Lake City Directory*, 1920

Notes: Each panel shows the top quarter of a page, below the advertising banner that heads every page. Panel (a) is from the alphabetical list of residents. Each entry gives the name, the occupation and employer in the publisher's abbreviations ("clk O S L" is a clerk at the Oregon Short Line railroad, "firmn S L Route" a fireman on the Salt Lake Route), and the home address, where "r" means resides, "b" boards and "rms" rooms; a widow is listed with her late husband's name in parentheses. Panel (b) is from the classified business directory, in which every firm appears under its line of trade with its address; the Continental Oil Co listed under Oil Dealers is the employer of the mechanic Geo C Lees in panel (a). Source: University of Utah, J. Willard Marriott Library, digital collections; scan at 300 dots per inch.

we define occupation code mappings separately for each occupation-employer industry-census era combination and take the modal census code within each cell. To avoid artificial changes in OCC1950 codes within persons created by changes in the modal occupational meaning over time, we assign codes by occupation-employer spells at the person level based on the spell's start date. For example, a man listed as a "driver" for the same employer from 1910

Table 1: Salt Lake City and Ogden city directories, 1900–1940: summary statistics

	Salt Lake City	Ogden
Listings per edition (mean)	70,556	19,542
# editions (years)	41	35
Occupation (% of listings)		
With a printed occupation	74.0	73.2
With an OCC1950 occupation code	73.6	73.0
With an employer	43.3	42.2
Residence precision (% of listings)		
Home address	69.4	69.8
Building	3.5	1.1
Street	2.5	2.3
Suburb or town	4.1	10.6
No address printed	4.1	5.8
Address not geocoded	16.4	10.4
Residence type (% of listings)		
Resides (r)	47.7	46.6
Householder (h)	15.2	16.2
Boards (b)	22.8	22.7
Rooms (rms)	7.8	7.3
No marker	2.3	1.4
Linking (% of listings)		
FamilySearch profile	62.0	64.0
1900 census	37.7	39.6
1910 census	46.0	48.7
1920 census	48.3	52.1
1930 census	45.4	49.0
1940 census	39.6	43.2

Notes: Shares of an edition's listings (one row per person per edition, suburban listings included), averaged over the city's editions of 1900–1940; the first two rows are counts. Ogden lacks the editions of 1901, 1902, 1904, 1905, 1909 and 1937. An employer is a printed employer string. Residence precision is the level at which the home address is geocoded: a house-level point, a named building, a street or corner, or the town. Residence type is the marker printed before the address: before 1930 the books print 'r' (resides), 'b' (boards) or 'rms' (rooms); from the 1930 editions they print 'h' for the householder and 'r' for every other resident, and 'b' and 'rms' disappear. Census rows: the listing's FamilySearch profile is linked to a record in that census, in any state.

to 1930 keeps one code, the one current in 1910, even though the census's own word for a driver changed over those years.

Our analyses use both the three-digit OCC1950 code and its aggregation to the eleven IPUMS-defined major occupation groups. Appendix Table A1 lists the five most common occupations in each group. Measured on these same-year pairs, the directory code and the census code agree on the three-digit code 55 percent of the time and on the major group 65 percent. Appendix Figure A2 shows a confusion matrix of census and city directory major groupings, and Appendix Table A2 lists the forty most common strings with the code each receives and its agreement with the census. Occupational status is the LIDO score of Saavedra and Twinam (2020), an occupational income score for 1950 averaged over Utah cells for the occupation and male sex, used in logs.

2.2 Employers

Employers are printed for 58 to 59 percent of listings with an occupation and for 60 percent of the person-years in the analysis sample, most often for clerical workers and managers (83 and 75 percent of their person-years) and rarely for farmers (2 percent). Employer names require standardization in order to correctly place co-workers together at firms and to avoid inflating estimates of job switching. To account for spelling variants and employer name changes across editions, we match firms on name, business address, owner names, and employee rosters. Specifically, we consider firms to be the same if any of the following criteria are met: (1) they match exactly on name, (2) they match closely on name and have the same street address or owners, or (3) the old firm disappears and a new firm appears at the same address with most of the same employees. Each printed employer is then matched to a firm record of its own edition, which resolves abbreviations such as "A S & R Co" to the American Smelting & Refining Company. The method matches 87 percent of employer listings to a firm record and yields 69,759 employers in the Salt Lake City and Ogden books across all years (Appendix Figure A6).

We measure an employer's size by the number of listings that name it in a city and edition. Large firms, with 100 or more listed workers, account for 41 percent of the listings that name an employer, while small firms, with fewer than five, account for 16 percent. Firms are classified into the major industry groups of the 1950 Census classification, using the modal 1940 census industry of the employer's linked workers where two or more are observed (64 percent of the listed workers at firms); otherwise we use the firm's heading in the directory's classified section or its name. Firms employing 12 percent of these workers cannot be classified. Because firm addresses are geocoded, we also observe the distance from

home to work for the 46 percent of employer listings where both points are located; the median is two kilometers.

To interpret the employer field, we compare each man's listing in the 1940 directory with his record in the 1940 census. We define four directory-based employment statuses: "listed with an employer", "listed with an occupation but not employer", "listed with no occupation or employer", and "likely self-employed", where the latter are listings with no employer who are also named as owner, proprietor, or partner of a firm in the same edition. For men aged 18 to 64, we then cross-tabulate these directory statuses with census employment statuses (Appendix Figure A7). Listings with an employer are nearly always identified as wage or salary employees in the census (89.4 percent). Listings with an occupation but not an employer were recorded as employed in the census only 55 percent of the time, while 14.1 percent were unemployed and 13.9 percent were on public emergency work. Thus, we interpret this group as people with a trade, but with less labor market attachment. Men we identify as likely self-employed in the city directories are usually self-employed in the census (67 percent). Finally, listings without an occupation map to census employment statuses differently depending on age. For men aged 18 to 24, 44 percent were in school, while 31 to 36 percent of those aged 35 to 54 were self-employed, and 32 percent of those aged 55 to 64 were out of the labor force for reasons other than school.

2.3 Addresses

Like many old Utah cities, SLC and Ogden are laid out on a grid, with street and house numbering indicating distance and direction from the city center. The earliest directories described residences using approximate locations along streets and between cross-streets (e.g., "south side 3rd South between 8th and 9th East"). Beginning in 1885, directories printed modern house numbers, allowing us to match address strings to modern census maps. We geocode each listing's home address in three stages. First, the street token is matched to a canonical street, with the historical names mapped to their modern equivalents (East Temple to Main Street, 1st East to State Street; Morgan, 1959). Next, the house number is placed on that street using current address points provided by the Utah Geospatial Resource Center, or, where no point exists, by interpolation within the Census Bureau's TIGER address range for the block. Finally, an address that names only a building, a corner or a street is kept at that coarser precision.

Across editions, 96 percent of Salt Lake City listings and 94 percent of Ogden listings print an address, and 72 and 74 percent of those are placed at a house-level point; a further 11 and 15 percent are placed at a named building, a street or the town, and the rest are

not placed by the geocoder. Directories also printed residence-type codes that record the person's relation to the household: "r" resides, "b" boards, "rms" rooms, and from 1930 "h" for the householder (Appendix Figure A3). Before 1930 boarders and roomers are 42 percent of listings; from 1930 the books mark only the householder ("h") and every other resident ("r"). Half of all addresses carry a single listing in a given edition; the mean is 2.3 listings per address before 1930 and 2.0 from 1930, or 1.4 and 1.3 counting only the household's own listings (Appendix Figure A4).

2.4 Linking people across editions

Within each city, we match individuals across editions on name and address. To capture within-city moves, we allow unique matches between individuals with the same name but a different address as long as they have either the same employer or a household member who also has the same name across editions. At this stage we miss matches for people who simultaneously moved and changed their household structure or job, but we partially address this limitation using the genealogical linking described below.

We successfully link 63 percent of Salt Lake City listings and 66 percent of Ogden listings to a record in the next edition. Restricting to men aged 18–64 who are linked to a FamilySearch profile and list an occupation, the analysis sample (Section 2.6), these year-over-year linking rates increase to 84 and 85 percent (Appendix Table A3). Link rates are a function of both real turnover in the city's population and inconsistencies in how individuals are recorded. To benchmark our link rates against the census, we follow the men of the analysis sample listed in the directory of a census year for ten years, in the directory and in the census. Of these men, 56 percent in Salt Lake City and 55 percent in Ogden are listed in the same city's book ten years later, while 45 percent are enumerated in the same city at the next census. Ten-year persistence in the directory is therefore at least as high as persistence in the city according to the census.

2.5 Linking to FamilySearch and the census

After merging city directory people across editions, we match each individual on name and location to a contemporaneous adult in the FamilySearch Family Tree, through which we link individuals to their census records.[3] FamilySearch is a wiki-style genealogical website that includes demographic information, family linkages, and attached census records for millions

[3]We are currently waiting on access to the restricted complete-count census files that will allow us to match city directory people directly with census records on name and address, without needing to go through FamilySearch. This will reach the residents who have no FamilySearch profile, a minority in Salt Lake City, and will adjudicate between candidate matches by address.

of people worldwide (Buckles et al., 2023; Price et al., 2021). Coverage is especially high in Utah, where the company is based, and among members of the LDS Church, which owns the company.

Between 57 and 70 percent of listings carry a FamilySearch link in every edition, and the share rises over the period as the tree's coverage of the 1880–1910 birth cohorts is dense (Appendix Figure A1). Using these links, between 32 and 41 percent of all adults enumerated in the census in SLC and Ogden, and between 45 and 65 percent of men aged 18 to 64, are matched to a listing in the directory that year (Appendix Table A4). Coverage of city directories varies by individual characteristic and is much higher for men than women. Appendix Figure A5 shows that coverage is lowest for men who were not heads of their own households, 14 percent for boarders and other non-relatives of the head and 4 percent for institutional inmates, compared with 63 percent for heads. It is also lower for the foreign-born (39 percent compared with 55 to 58 percent for the native-born), for non-white men (10 percent), for men under 25, and for laborers and service workers (43 to 45 percent) than for clerical, professional and managerial workers (62 to 66 percent).

2.6 Analysis sample

Our analyses use a panel of men aged 18–64. To measure age, we further restrict to men for whom we have a linked FamilySearch profile, which provides date of birth. Of the 3.58 million listings in the two series, 2.24 million are linked to a FamilySearch profile, and 1.44 million of these are men aged 18 to 64. A man appears in 7.6 editions on average (median 5), and the 46 percent of men who appear in six or more editions account for 83 percent of person-years.

3 Firms in Pre-War Utah

The early twentieth century saw a shift from agriculture and sole-proprietorship towards wage labor. Among men aged 18 to 64 in the labor force, the share working in agriculture fell from 38 percent in 1900 to 20 percent in 1940, and the share who were self-employed from 32 percent in 1910 to 22 percent in 1940 (Ruggles et al., 2025). Increasingly, employment was organized in large corporations run by salaried managers (Chandler, 1977; Jacoby, 1985). Yet, firms still offered few benefits and job turnover was high (Jacoby and Sharma, 1992; Owen, 1995), and many Americans still held the ideal of the independent proprietor (Blackford, 2003; Mills, 1951). It is therefore an open question how much it mattered which firm a man worked for, and how much he lost when that firm closed.

In this section we describe the role of firms in the pre-war Utah urban economy, focusing on the following four questions. What did firms look like in terms of size, structure, and market concentration? What kinds of people worked at a firm instead of being self-employed or unemployed? How did turnover and advancement unfold over a worker's career? How much did wages and non-wage amenities vary across firms?

3.1 Characterizing firms

Between 1900 and 1940, the total number of firms identified in the directories as operating in SLC and Ogden grew from 1,160 to 3,619. Most firms (56 percent) were small, employing only two to four employees, but the distribution of firm size was highly skewed such that 40 percent of workers were at the largest 2 percent of firms with 100 or more employees (Figure 2, panel A). Firm size varied sharply by industry with 88 percent of workers in mining and 83 percent in transportation, communication and utilities listed at firms of 100 or more, compared with only 2 percent in finance, insurance and real estate (panel B).

The largest employers were the railroads, the mining and smelting companies, the utilities, and a few large retailers (Appendix Table A5). In 1910 the Oregon Short Line and the Denver & Rio Grande Western railroad companies each employed more than 1,000 workers in Salt Lake City, and the Southern Pacific railroad company employed 709 in Ogden. By 1920 Utah Copper and American Smelting & Refining had emerged as a similarly large employer, and by 1940, Utah Copper was the largest employer of the two cities, with 2,147 listed workers. Meanwhile, Zion's Cooperative Mercantile Institution (ZCMI), the department store organized by Brigham Young and the leaders of the LDS Church in 1868 (Arrington, 1958), and the Mountain States Telephone & Telegraph Company were among the ten largest employers in every census year. Employment nonetheless became less concentrated as the cities grew as the employment share at the ten largest employers fell from 29 percent in 1900 to 20 percent in 1940 (Figure 2, panel C).

Labor-market concentration, measured by the employer Herfindahl–Hirschman index (HHI) within each occupation code, similarly fell across time. Between 1900 and 1940, the share of workers in a highly concentrated market with an HHI above 2,500[4] fell from 20 percent to 11 percent (Figure 2, panel D). Workers in highly specialized occupations faced more concentrated labor markets. Between 82 and 100 percent of the railroad brakemen, switchmen and locomotive engineers, 90 percent of the telephone operators and 98 percent of the streetcar motormen worked in highly concentrated markets, as did 37 percent of workers in public administration and 34 percent in transportation, communication and utilities,

[4]Federal merger guidelines use an HHI of 2,500 as the threshold for defining a highly concentrated market.

compared with only 2 percent in trade. Workers at the largest firms were the most exposed (Appendix Figure A8), but 89 percent had at least five other employers of their occupation in the same city, so the majority of workers had options if their employer closed. The exceptions were men in the specialized trades of the railroads, the utilities and the smelters, whose large employers rarely closed (Section 4.1).

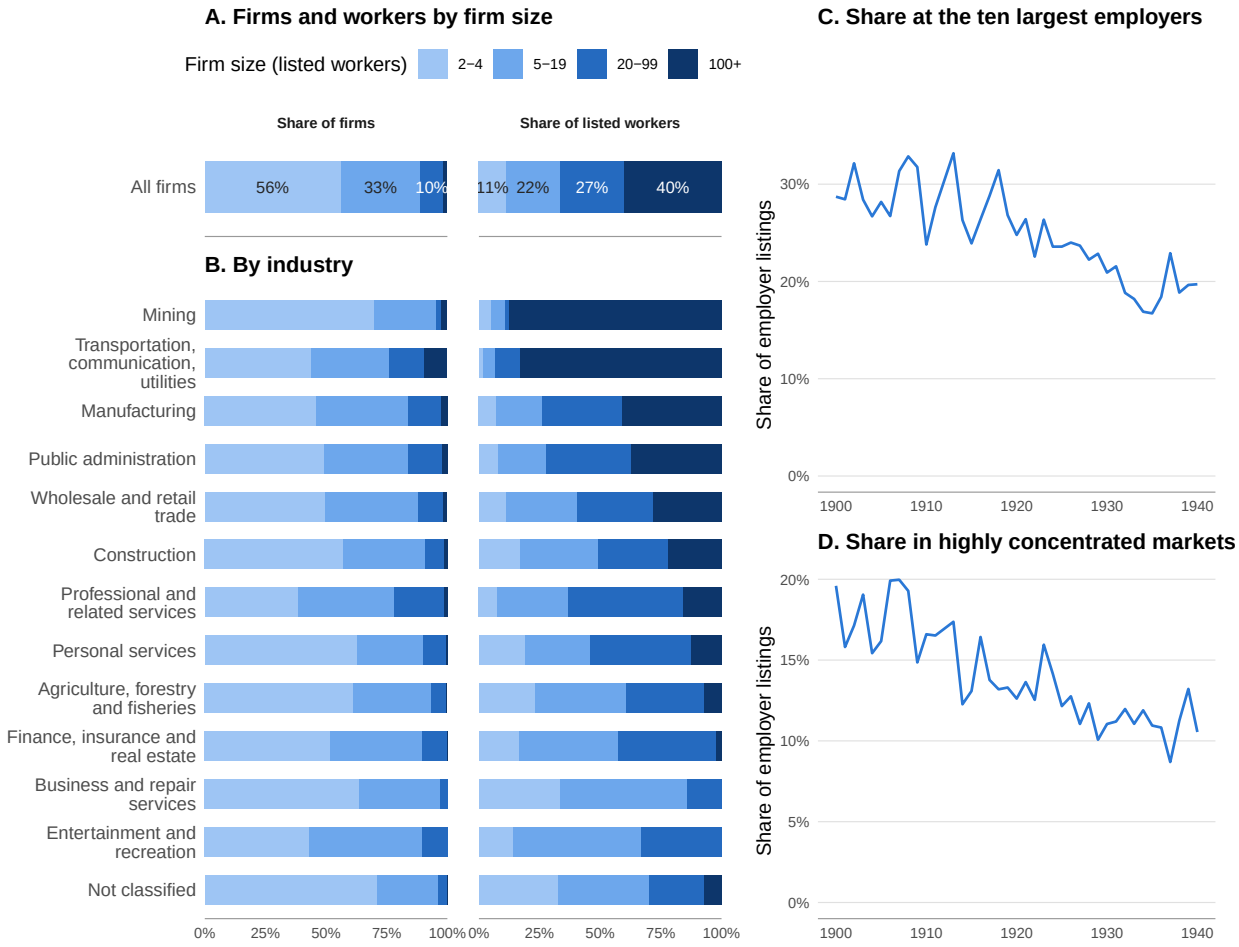


Figure 2: Firm size and the concentration of employment, 1900–1940

Notes: Salt Lake City and Ogden directories, 1900–1940. Panel A compares the share of firms belonging to each firm size category with the share of workers employed by firms in that category. Panel B splits this comparison by industry (1950 Census industry major groups). Panel C shows the share of employed workers listed at one of the year's ten largest employers. Panel D shows the share of workers with an employer Herfindahl–Hirschman index above 2,500.

3.2 Who worked for a firm

Most working-aged men in pre-war SLC and Ogden worked for wages, with the fraction rising from 72 to 77 percent between 1910 and 1940.[5] Our measure of firm employment, the fraction of directory workers with an employer, is slightly lower at 60 percent, but is highly correlated. As discussed in Section 2.2, 89.4 percent of the men listed with an employer in the 1940 directory worked for wages or a salary in the 1940 census.

Figure 3 shows which characteristics predict being listed with an employer, from a linear probability model with city-by-year fixed effects. The probability falls with age, by 17.9 percentage points from ages 18–24 to 55–64, which could reflect unemployment rising with age or an increased propensity by self-employed among older men. Employment is 6.4 points higher for married men.

Foreign-born men were 1.8 percentage points less likely to be listed with an employer while second-generation men were no different from men with U.S.-born parents. Both Salt Lake City and Ogden were founded by members of the Church of Jesus Christ of Latter-day Saints (LDS pioneers), but we find that their descendants who stayed in the two cities were 4 percentage points less likely to be employed at a firm.

Employment appears to transmit intergenerationally. We find that men whose father had been listed with an employer before they turned 18 were 8.2 points more likely to be listed with one themself. Furthermore, 14.8 percent of the employer-years of sons were at an employer for which their father had worked (Appendix Figure A9).[6] The share is highest for young men, 18 percent at age 20, and falls steadily with age, to 14 percent at 30 and 10 percent at 60, as sons moved on from the firms that first hired them. It also declined over the period, from 19.8 percent in the 1900s to 13.6 percent in the 1930s.

Finally, men who lived further from the central business district were less likely to work for a firm, with a decline of 9.4 points for men located four or more kilometers from Temple Square or downtown Ogden.

3.3 How firms shaped careers

Young men changed occupations and employers often, but stabilized as they aged (Figure 4, panel A). At age 20, 42.5 percent of men with an occupation in two consecutive years changed their occupation code, and 31.1 percent of men with an employer in both years

[5]The fraction of men working for wages is constructed from the census using the unemployment questions of 1910 , 1930, and 1940. In 1940 it includes men on public emergency work.

[6]If sons were assigned to firms at random, only 1.4 percent would work for an employer where their father had worked.

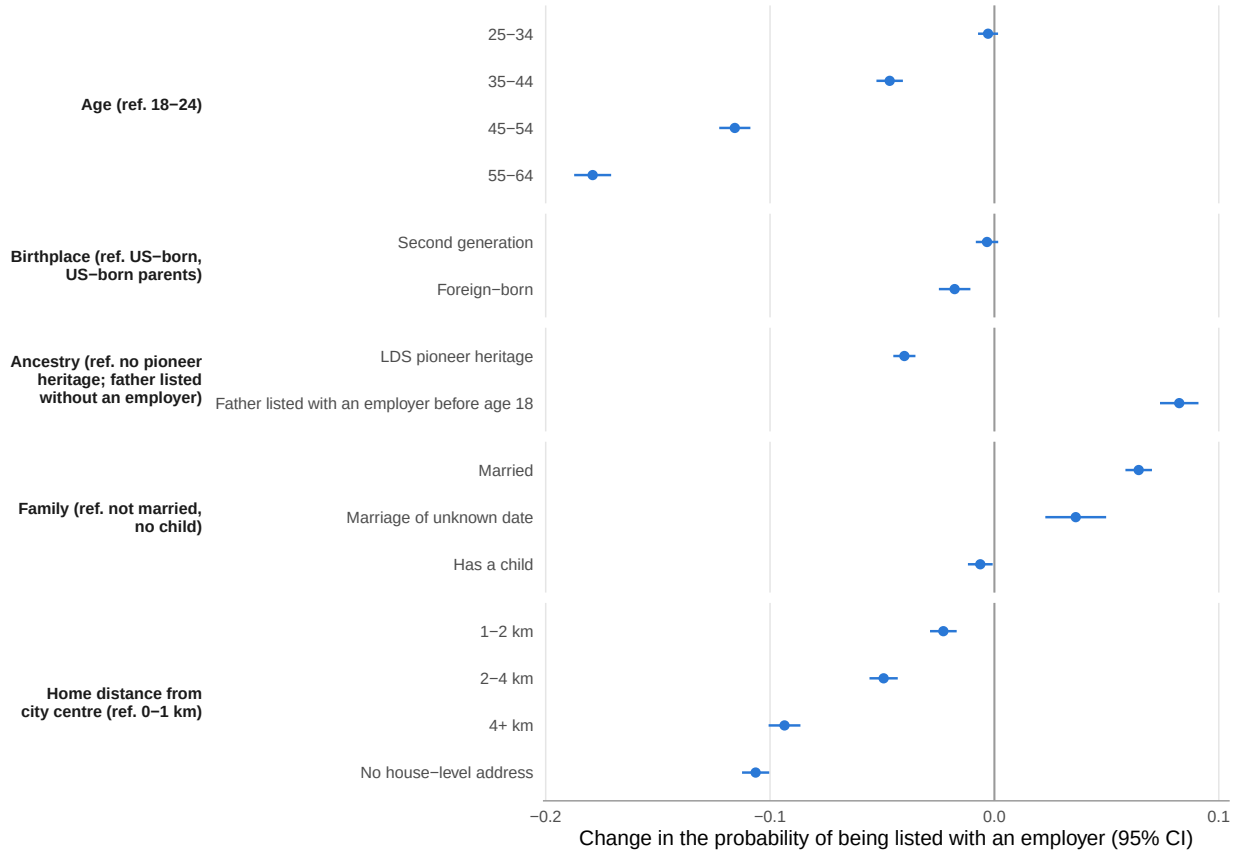


Figure 3: Who worked for a firm: predictors of being listed with an employer

Notes: Coefficients and 95 percent confidence intervals from a linear probability model of being listed with an employer, with city-by-year fixed effects; standard errors clustered by man. Analysis-sample person-years, 1900–1940 (1,078,821 person-years, 142,429 men). Birthplace and parents' birthplace are from FamilySearch; second generation: born in the United States with a foreign-born parent. LDS pioneer heritage: the man, a parent or a grandparent listed in Mormon pioneer overland and transatlantic registries. Father listed with an employer: in any of the four cities' directories before the son turned 18. Distance: from Temple Square (Salt Lake City) or 25th Street and Washington Boulevard (Ogden) to the home, for addresses geocoded at the house level.

changed employer, while at 40 the rates were 23.2 and 18.1 percent, and at 60 they were 19.2 and 14.6 percent.

Until age 40, men advanced by switching towards higher paying occupations. Following the same men over time, occupational status rose by 7.4 log points between ages 23–27 and 43–47 and then stayed roughly flat, 6.0 log points above its level at 23–27 at ages 63–64 (panel D). Occupational advancement was about twice as likely to occur for workers as they switched employers than for those who stayed (panels B and C). However, because most workers remained with their employer year-to-year, most occupational advancement took place within firms: 52 percent of the year-to-year moves up of men with an employer in both years at age 20, 67 percent at age 50 and 77 percent at age 60 occurred without a

change of employer.

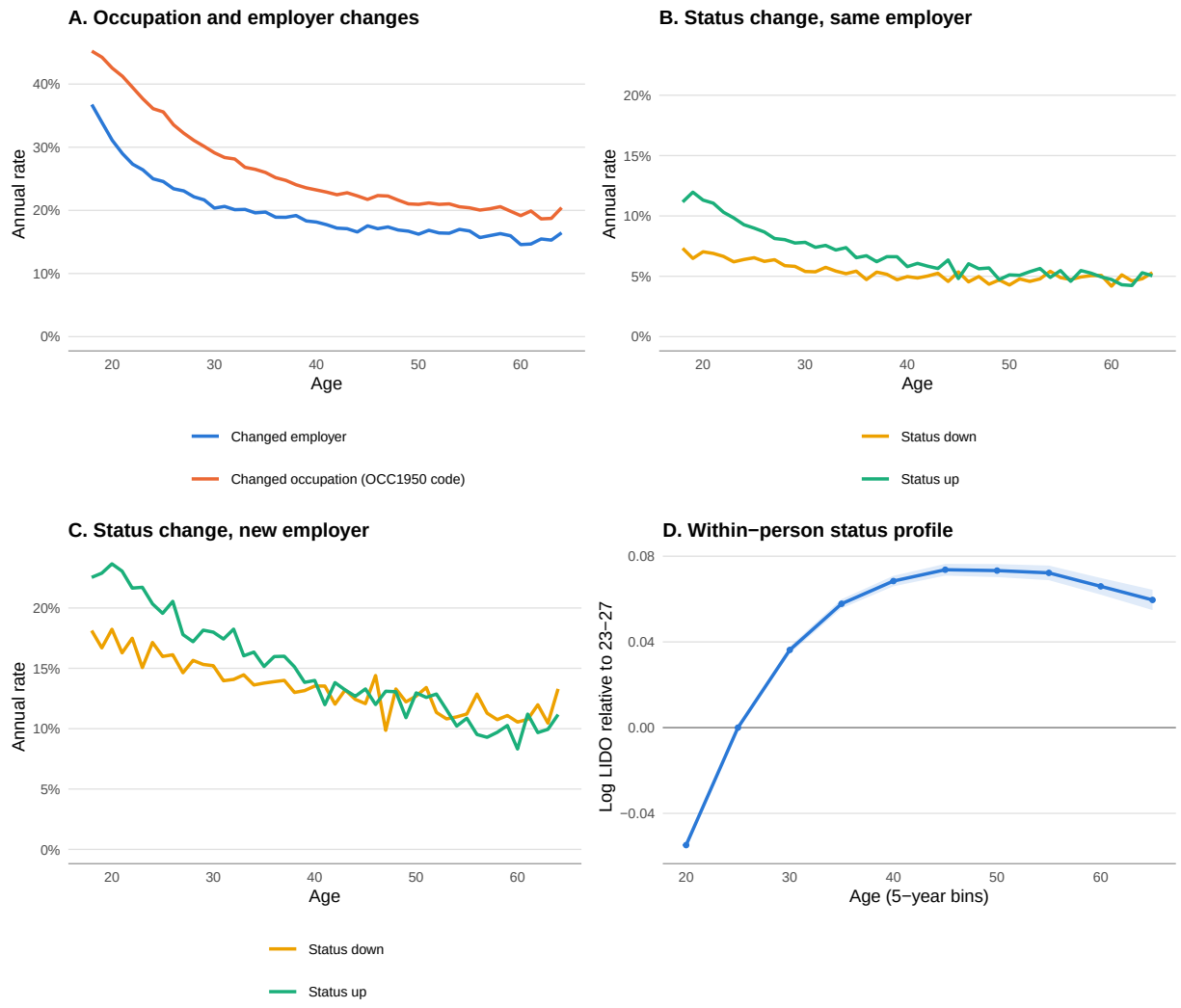


Figure 4: Occupational change and occupational status over the life cycle

Notes: Analysis-sample men aged 18–64, 1900–1940. Panel A: among men with an occupation in years t and $t+1$, the share whose OCC1950 code changes; among men with an employer in both years, the share whose employer changes. Panels B and C: among men with an occupation and an employer in both years, the share whose log LIDO rises or falls by more than 0.1, for men who kept their employer (B) and men who changed it (C). Panel D: coefficients on five-year age bins (reference 23–27) from a regression of log LIDO with person fixed effects; 95 percent confidence intervals clustered by man.

Tenure with a firm was also rewarded in pay. Table 2 links the employer in the 1940 directory to the wages and hours that the same men report in the 1940 census. Within an occupation, the log weekly wage rises with each additional year at the firm, increasing by 0.15 to 0.17 for workers with five or more years at the company.

Altogether, these patterns fit what historians have found by analyzing firm records.

Alongside many short jobs, a core of long-tenure jobs existed before the First World War and grew after it, as large employers built personnel departments and promoted from within (Carter and Savoca, 1990; Jacoby, 1985; Jacoby and Sharma, 1992). The city directories show the same features across the whole labor market of two cities, including the small firms that firm records rarely cover. During our period, much career advancement transpired within firms, through internal promotion and pay that rose with tenure, a pattern the modern literature attributes to firm-specific skills (Topel, 1991).

Table 2: Employer size, tenure and 1940 wages and hours

	(1) Log annual wage	(2) Log weekly wage	(3) Hours last week
Employer size 5-19	0.040 (0.018)	0.058 (0.015)	-0.945 (0.323)
Employer size 20-99	0.064 (0.019)	0.077 (0.015)	-1.715 (0.339)
Employer size 100+	0.052 (0.023)	0.084 (0.020)	-2.041 (0.436)
Editions with the employer 2	0.240 (0.016)	0.062 (0.012)	0.685 (0.260)
Editions with the employer 3-4	0.271 (0.017)	0.097 (0.011)	0.182 (0.247)
Editions with the employer 5-9	0.323 (0.017)	0.152 (0.012)	0.110 (0.302)
Editions with the employer 10+	0.350 (0.018)	0.166 (0.014)	-0.030 (0.349)
Occupation fixed effects	X	X	X
Mean of outcome	7.25	3.41	45.7
Men	18,994	18,994	14,523

Notes: Men aged 18–64 in the 1940 census of Salt Lake City and Ogden, linked to FamilySearch, working for wages with positive wage income and weeks worked, and listed with an employer in the 1940 directory; column (3) men at work in the census week. Wages are 1939 wage and salary income; weekly wage = annual wage divided by weeks worked. Employer size: listings with an occupation naming the employer in the 1940 directory (reference 1–4). Editions with the employer: consecutive directory editions naming the same employer, ending in 1940 (reference 1; spells begun by 1898 are in 10+). Occupation fixed effects: the OCC1950 code of the 1940 census occupation. All columns control for age bands and city. Standard errors clustered by employer.

3.4 Differences between firms

Firms differed in what they paid for the same work. The combination of job tenure, age, and city and occupation fixed effects explain 44 percent of the variation in log annual wages. Adding employer fixed effects explains an additional 8 percent of variation (Appendix Figure A10).

Beyond pay, firms also differed in their non-wage amenities. To measure how much workers valued each firm, we follow Sorkin (2018) ranking firms using a revealed preference measure based on employee's voluntary moves between firms. An employee who moves from one firm to another reveals a preference for the new firm, with moves from high value firms conveying more information about the destination firm quality than moves from low value firms. Involuntary separations due to layoffs do not convey worker revealed preference. We identify such exogenous separations as separations in excess of the usual rate at shrinking firms, correcting for differences in firm size.

We estimate firm value rankings on the 2,307 firms with ten or more employees in at least one edition who form a strongly connected set of moves between firms. These firms employ 77 percent of the person-years with an employer in the analysis sample. Within this set, there are 31,106 moves between firms and 19,249 moves to or from being listed without an employer. Values estimated from two random halves of the men correlate 0.67, and 58 percent of the moves between firms go to a higher-value firm, compared with 51 percent to a firm with a higher status premium and 49 percent to a larger firm (Appendix Figure A11).

Figure 5 shows characteristics of firms that workers preferred. Workers show no preference over firm size and but placed lower value on the firms that closed during the Great Depression (0.67 standard deviations lower value). Industry matters a lot, with workers strongly preferring jobs at firms classified as public administration (1.43 standard deviations above trade), followed by finance, insurance, and real estate (0.73), and professional services (0.58). Personal services and entertainment rank lowest at -0.43 and -0.76 standard deviations below trade.

Among the large employers, the post offices of the two cities, the Railway Mail Service, the Federal Reserve Bank, the Veterans Administration and the Salt Lake City police department rank at the top, and hospitals, hotels, a department store, a smelter and the gas and telephone companies near the bottom. Firms that employed men in higher-status occupations were not valued more (a correlation of 0.06 between value and the status premium), but firms that paid more were. In 1940, a firm's value and its wage premium within occupation correlate 0.42 after correcting for the noise in both measures.

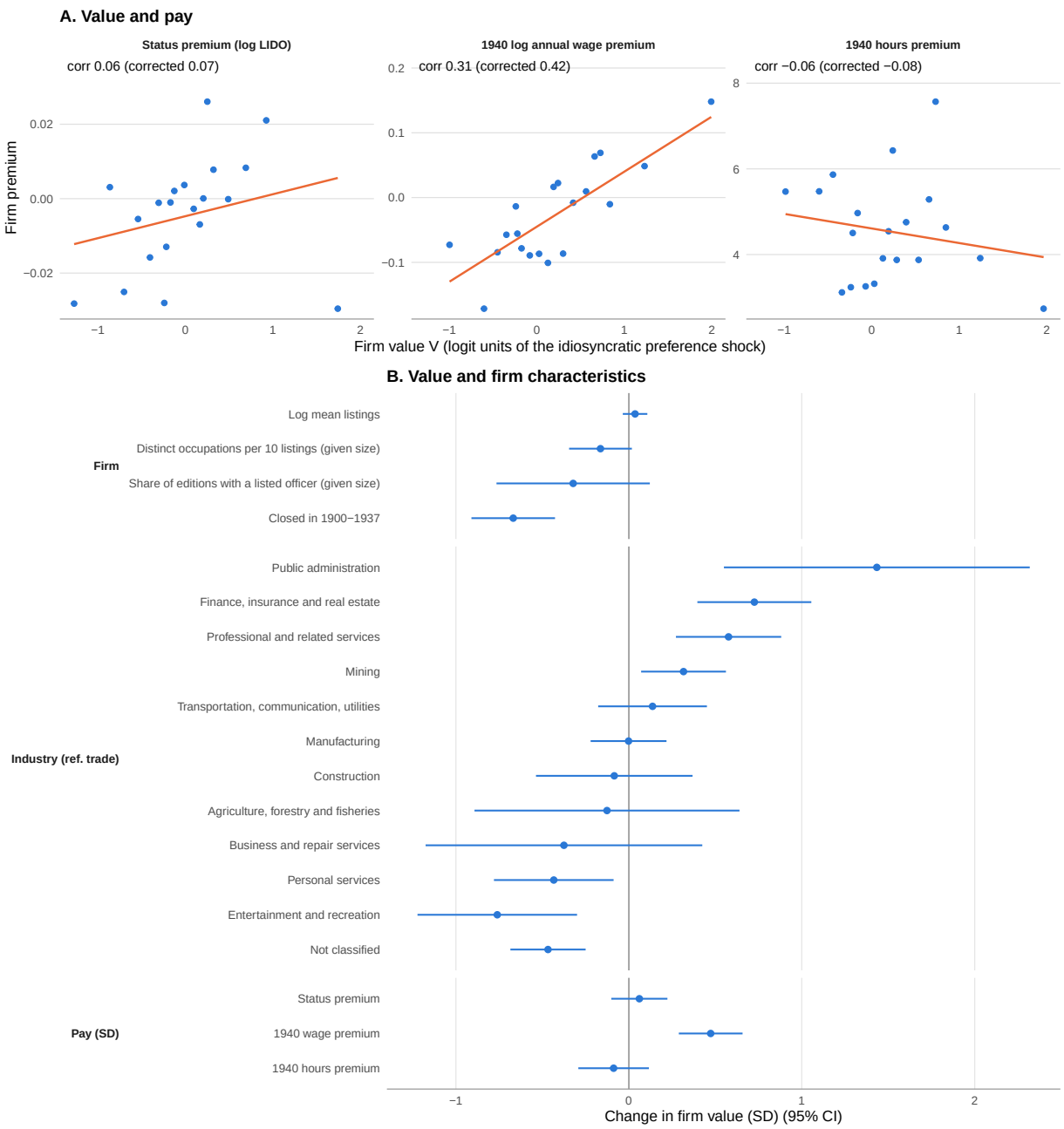


Figure 5: Which firms did workers prefer? Firm values from revealed preference

Notes: Firm values V estimated as in Sorkin (2018), in units of the standard deviation of the idiosyncratic preference shock, from the moves of analysis-sample men between consecutive editions, 1900–1940: 2,307 firms (a harmonized employer in a city) with ten or more listings in some edition that form a strongly connected set and hire men without an employer. Panel A: binned scatter (20 bins weighted by person-years) of firm premia against value. Status premium: employer fixed effect in log LIDO with person and year fixed effects. 1940 wage and hours premia: employer fixed effects in the regressions of Table 2 with OCC1950-code fixed effects, firms with four or more such men. Corrected correlations divide by the square root of the split-sample reliabilities of value and premium. Panel B: value in standard deviations regressed on each characteristic separately weighted by person-years.

4 Research Design

As discussed in Section 3, firms played an important role in pre-war Utah's urban economies. Most workers were employed by and advanced over their career at a firm, and firms varied in the wages they paid and amenities they provided to workers in the same occupation. To further understand the importance of firms, we next study the impact of firm closures on worker outcomes. We study every closure of a firm with ten or more listings between 1901 and 1938, and compare its employees to men at surviving firms who had followed similar career trajectories.

4.1 How often firms closed

We define a firm closure as occurring when a firm existing in one directory is then absent from the next three editions. We choose three editions because it balances two aspects of our data. First, because directories firm listings were incomplete, sometimes leaving out a firm in one year, only to add it back the next, defining firm closures using only one absence would conflate true closures with missing data. Second, our data is right censored at 1940 meaning that requiring a longer window of absence would shorten the window over which we can measure firm closures. A window of three years, allows us to measure firm closures through 1937, capturing most of the Great Depression. Our closure rate also leaves out government and public organizations as well as firms whose workers reappear together at another employer, indicating a merger, acquisition, or renaming, rather than a closure.

Using this measure, an average of 17.2 percent of firms closed each year, with closures concentrated among young and small firms. Of firms in their first edition, 48 percent of those with two to four workers closed, compared with 25 percent of those with five to nineteen workers and 10 percent of those with 20 to 99 workers. Additionally, the hazard rate of closing dropped with each year a firm survived, stabilizing after about five years (Appendix Figure A12).

Closure rates fluctuated across time, spiking around the Panic of 1907 and the first few years of the Great Depression. Between 1907 and 1908 the closure rate rose from 18 percent to 23 percent and remained high through 1911. Most of this increase was driven by small firms in trade, personal services, and entertainment. The Great Depression saw a similar jump from 16 percent a year in 1925–1929 to 21 percent in 1932 (Figure 6).

Firms did not fail at random, and those that closed differed from those that survived. Appendix Figure A14 compares, for every base edition b, the firms with at least ten listed employees that closed after b with those that survived, pooled across all editions. Closing

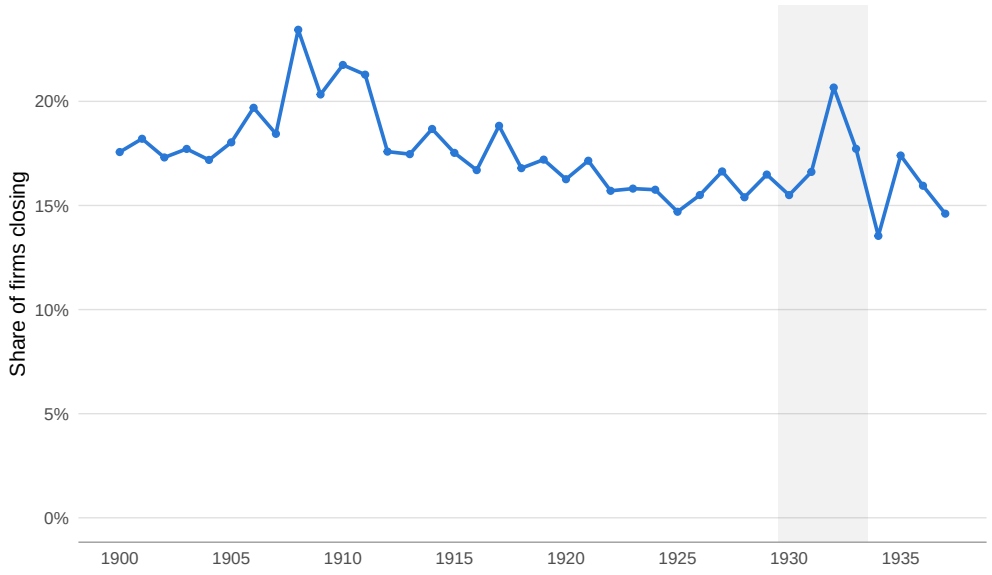


Figure 6: Annual closure rate of firms, 1900–1937

Notes: Salt Lake City and Ogden pooled. A firm is defined as closing in year t if it is absent from the city's next three editions. Government and public bodies are left out, as are mergers, acquisitions or renamings. Shading marks the early Great Depression years (1930–1933).

firms were smaller (15 listed employees vs. 28), newer (5 years old vs.10.6 editions) and less likely to print a corporate officer (29 vs. 48 percent). They were also lower-valued firms as measured through voluntary employee moves in Section 3, by 0.8 standard deviations. On industry, they were more often unclassified small firms (39 vs. 11 percent) or mining firms (3.5 vs. 0.5 percent), and less often in trade (22 vs. 36 percent), manufacturing (14 vs. 22 percent), and finance (2 vs. 10 percent). Closures occurred suddenly with no evidence for mass layoffs in the years preceding (Appendix Figure A13).

4.2 Treated and comparison men

Treatment is defined at the level of the employer-year. Treated individuals are the 3,297 men listed at 508 closing firms in the year it closed, which we call the base year and denote b. The comparison group consists of 68,016 men listed at firms that survived and maintained at least 75 percent of their workforce for at least eight editions after b, which is the length over we which we follow its workers. Relative to men in our comparison group, men at closing firms were slightly younger (34.2 vs. 35.9 years), had been with their employer for half as long (1.8 vs. 3.6 years), worked at much smaller firms, and were observed less often in the five year period leading up to b (Appendix Figure A15. These patterns match the differences between failed and surviving firms discussed above.

To address differences between treated and control men, we match men on pre-treatment career paths using a propensity score of being at a closing employer, estimated separately for each city-base year. We use a logit with the following predicators: age (quadratic), tenure with the employer, occupational status (LIDO score) in b, occupation group, the employer's industry and size, and indicators for being listed, listed with an employer and listed with an occupation in each of the five directory editions before b. Each treated man is matched with up to three comparison men of the same occupation group whose scores lie within 0.2 standard deviations of his on the logit scale. We match 3,038 of the 3,297 treated men to 4,934 comparison men. After matching, the standardized difference between the groups is below 0.06 for every matched variable, compared with up to 0.74 before matching.

Our study window includes the Great Depression and one may wonder whether effects from firm closures evolved differently during this period compared with other eras. In particular, because the Depression was characterized by so much economic and policy change, it is instructive to study effects as they unfolded over calendar, in addition to event-study, years. For this analysis, we pool 89 firms that existed in 1929 but closed at some point during 1930–1933 and match them to 319 surviving firms. The treatment group consists of 804 men at the 89 firms that close who we match on 1920–1928 career paths and 1929 characteristics with 1,080 comparison men (Appendix Table A6).

4.3 Estimating equation

We follow every man in each edition from five years before b to eight year after, whether or not he is listed in all editions. For years he is not listed, outcomes are set to zero unless he is dead, in which case they are dropped. Outcomes measured using the directories include whether the man is listed in that year's directory, whether he is listed with an occupation, whether he is listed without an occupation, whether he is listed with an employer, and whether he is still at his year b employer. For men listed with an occupation, we also measure the change in log occupational status since b. In addition, we use three outcomes derived from FamilySearch linkages: whether the man is observed living outside the county, whether he is married, and his total number of kids.

Using the terminology of Cengiz et al. (2019), we call a unique city-base year combination a stack, denoted by s. For man i in stack s and edition t we estimate

$$y_{ist} = \alpha_{is} + \lambda_{st} + \sum_{\tau=-5,\,\tau\neq 0}^{8} \beta_\tau \, \mathbf{1}[t - b_s = \tau] \, D_{is} + \varepsilon_{ist}, \tag{1}$$

where y_{ist} is one of the outcomes above and D_{is} equals one if man i's employer in stack s closes

and zero if it survives. The person-by-stack fixed effect α_{is} absorbs every fixed difference between men, and the year-by-stack fixed effect λ_{st} absorbs the common path of the stack's men over time. Each coefficient β_τ is the difference between treated and comparison men in the change in y τ years after b (event time). Standard errors are clustered by base-year employer, which is the level at which treatment is assigned. Because comparison men can be matched to multiple treated men, we assign them a weight equal to the inverse of the number of treated men with whom he matched.

Because each closure is compared only with surviving employers of the same editions, our estimates do not suffer from the problems of two-way fixed-effects estimators with staggered treatment timing, in which men treated earlier serve as controls for men treated later.

4.4 Identifying assumptions

Estimates of β_τ in years following b represent the causal effect of a firm closure on worker outcomes τ years out under the standard three event-study identifying assumptions: parallel trends, no anticipation, and SUTVA. We discuss each and assess its plausibility below.

The parallel trends assumption requires that, had a man's employer survived, his outcomes would have evolved similarly to his matched comparison men. Absent matching, this is unlikely to hold, since employees at failing firms are systematically different and have different prior career trajectories than employees at surviving firms. By matching treated individuals to comparison men with similar career paths, parallel trends is much more likely to hold, and we show that pre-trends are flat even for outcomes not used in matching.

Second, the assumption of no anticipation requires that firm closure does not change behavior in the pre-periods. If the ablest workers at a failing firm were able to anticipate its closure and chose to leave ahead of time, one might worry that the remaining workers observed in year b were negatively selected. Reassuringly, however, we find no evidence for anticipatory firm switching whether on the part of workers or in the layoff behavior of firms since employment counts remain steady in the years leading up to a closure (Appendix Figure A13).

Third, SUTVA requires no spillover effects of firm closures on comparison workers at surviving firms. This is unlikely to be true, as the failure of one firm likely affected the success of competitor firms and the job options of their workers. Instead, we interpret our estimates as the effect of an individual belonging to a failing firm rather than the surviving firm of his matched comparisons.

5 Results

5.1 Effects of firm closure

Figure 7 plots the coefficients β_τ of equation (1) for six directory outcomes and three outcomes from the family records of FamilySearch, described below. Pre-trends are similar between treated and comparison men for all outcomes. At a statistical level, our first stage outcome—whether a man is still listed at his base-year employer—has a significant pre-trend, however, it is small in magnitude relative to the effect. Its p-value for a joint test that all five pre-period coefficients are zero is 0.002. P-values for all other outcomes are well above conventional significance levels: listed in a directory ($p = 0.67$), listed with an occupation ($p = 0.63$), listed with an employer ($p = 0.75$), listed without an occupation ($p = 1.00$), occupational status ($p = 0.14$), moved away from county ($p = 0.55$), marriage ($p = 0.22$) and fertility ($p = 0.30$).

We first examine the effect of firm closure on whether a man is still at his base-year employer. By construction, no treated man was still at his base-year employer after it closed. However, as established in Section 3, employer turnover was high, and about half of the comparison men had also left their employer within a year. Thus, one year after b, a man whose employer closed was 52.5 percentage points less likely than his matched comparison men to be listed with his base-year employer, with the gap narrowing to 21.6 percentage points after eight years as more comparison men left their employers over time.

Men responded to the loss of their employer in a variety of ways, and many effects decline but do not fully disappear over the course of eight years. A substantial fraction of men moved away. Firm closure reduced the probability that a man was listed in either city's directory by about ten percentage points (evidence of leaving his city) and, from the third year on, increased the probability that his last known residence was outside his county by three points. Most of those who left their county left Utah altogether (2.3 points after eight editions; Appendix Figure A16).

Those who stayed in the city were divided between unemployment, weak labor force attachment, and lower-status work. Among those who remained listed, an additional five percentage points of men were listed without an occupation or employer the year after their firm closed, an effect that faded out after about six years, and another fourteen percentage points had an occupation but no employer, an effect that fell to two points by the eighth year. Additionally, those who found employment did so in lower paying jobs scoring three percentage points lower in occupational status. We detect no significant effects on marriage and fertility, however, it is worth noting that our estimates are imprecise.

Appendix Figure A16 shows effects for several additional outcomes. Displaced men, as well as their wives and children, were no more likely to die, and their wives did not respond to their job loss by entering the work force.

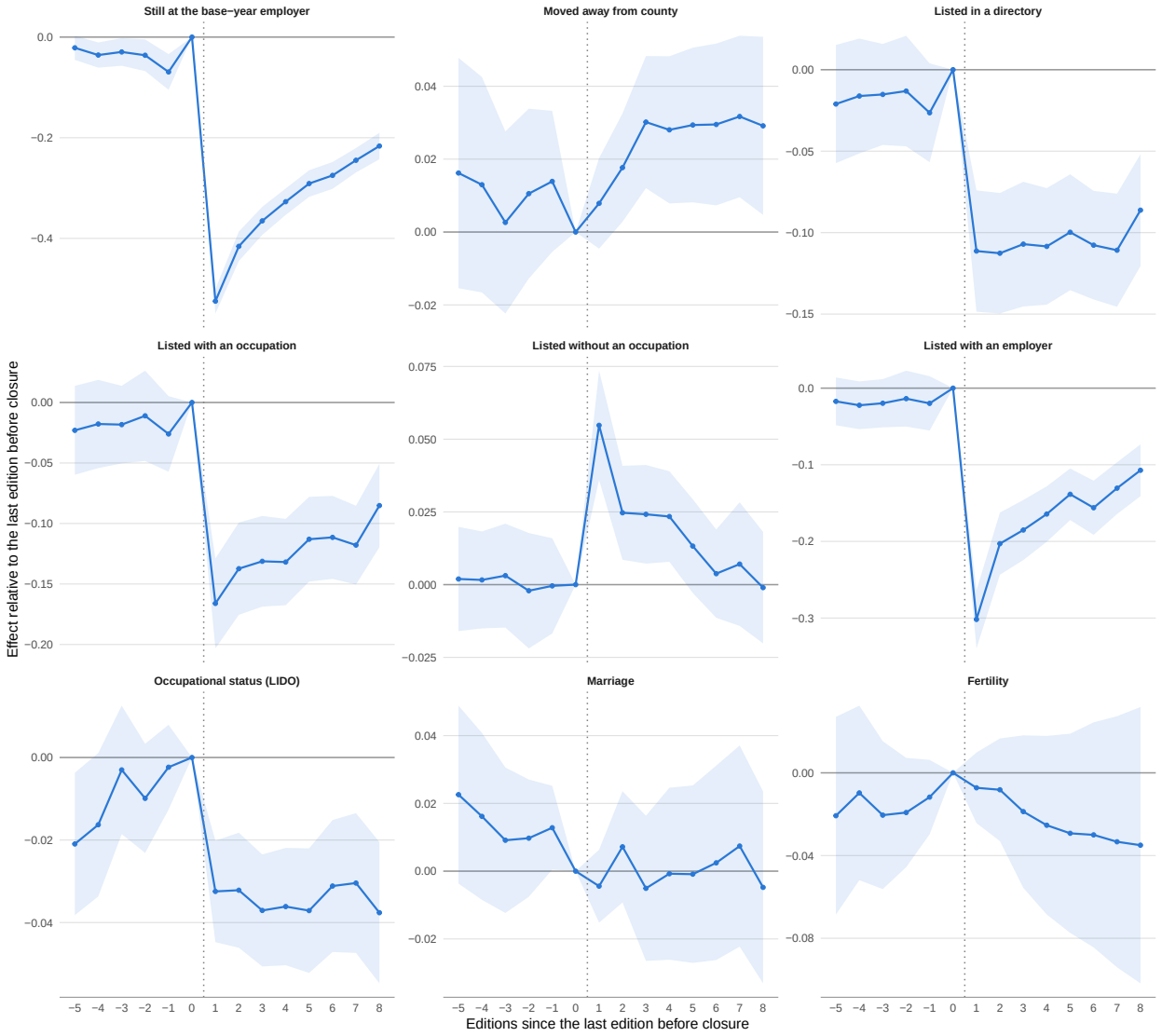


Figure 7: Effects of firm closure, 1901–1938

Notes: Coefficients β_τ of equation (1) and 95 percent confidence intervals, in editions relative to the last edition before closure (b). Stacks are a city and a base edition b = 1900–1937. Regressions include 3,038 closing-firm men (treated) matched to 4,934 comparison man-editions in surviving firms. Directory outcomes are zero when the man is not listed, and person-years after a recorded death are dropped. Occupational status is log LIDO minus its value in b, for men listed with an occupation. Moved away from county: the man's last known residence, the place of his most recent directory listing or located FamilySearch or census record, is outside Salt Lake County (Salt Lake City stacks) or Weber County (Ogden stacks). Marriage: a dated FamilySearch marriage with no evidence of having ended by the year. Fertility: the number of children born to the man by the year.

5.2 Were Depression closures worse?

To study whether closures during the Great Depression were worse for workers than closures in other years, we re-estimate the event studies separately for firms that closed in 1930–1933 and for firms that closed in all other years (Figure 8). Men displaced in the Depression fared much worse in the short run. One year after closure, they were 43 percentage points less likely to be listed with an employer, compared with 28 points for men displaced in other years, and 20 percentage points less likely to be listed in a directory at all, compared with 9 points. Those who remained in an occupation lost ten percent in occupational status, five times the loss in other years. This is consistent with work on modern job loss which generally finds larger negative effects during recessions (Davis and von Wachter, 2011).

Over time the Depression gap closed. Eight years after closure, men displaced in the Depression were as likely as other displaced men to be listed with an employer, their excess loss in being listed at all had disappeared, and only the status loss remained larger, at six percent compared with three. Men displaced in the Depression were no more likely to move away, and we find no differences in marriage or fertility.

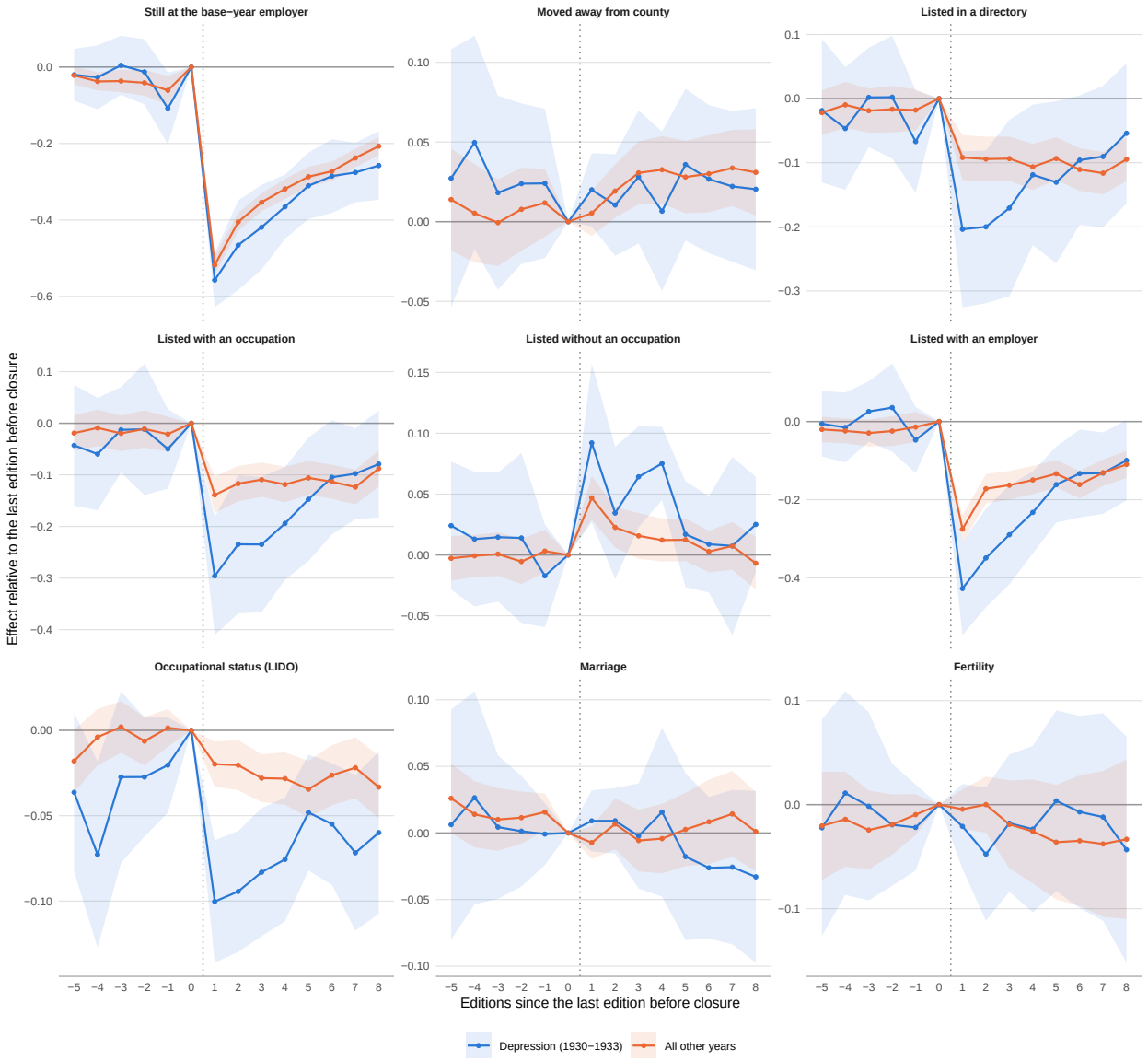


Figure 8: Effects of firm closure in the Depression and in other years

Notes: The outcomes and design of Figure 7, estimated separately for closures in the Depression (the city's first edition without the employer falls in 1930–1933: 525 closing-firm men at 68 employers in 7 stacks, matched to 732 comparison man-editions) and in all other years (2,513 men at 394 employers in 49 stacks, matched to 4,202).

6 Conclusion

In order to study the role of firms in the pre-war urban economy, we constructed a novel dataset from an underutilized source—U.S. city directories. City directories allowed us to measure each resident's occupation, employer, and address at an annual, rather than decennial, frequency. By transcribing and linking the Salt Lake City and Ogden directories

for 1900–1940, we built an annual linked employer-employee panel of 1.44 million person-years that follows men across editions, to their employers, and through FamilySearch to the censuses and to their families.

Using this newly constructed panel, we show that firms were central to careers during a time of high job turnover and before unemployment insurance and most benefits. About 60 percent of working-aged men were listed with an employer, and the largest 2 percent of firms employed 40 percent of them. Men advanced in occupational status until around age 40, and most of that advancement took place within firms, where pay also rose with tenure. Firms differed in the wages they paid for the same work and, judging by the revealed preferences of the workers who moved between them, in additional non-wage amenities.

Firm closures had significant negative impacts on their workers. Men whose employer closed were 30 percentage points less likely than matched comparison men to be listed with an employer a year later and still 11 points less likely eight years later. A tenth left the city, and those who remained were more often without steady work and in lower-status occupations, a status gap of 3 to 4 percent that did not close. In spite of this, we find no detrimental effects on marriage, fertility, or mortality. Closures in the early years of the Great Depression were far more damaging at first, but within eight years their effects had converged with those of closures in other years. In an era without unemployment insurance, the costs of displacement were as persistent as those documented in modern administrative data, and they took the form of lost attachment to steady employment and out-migration.

City directories survive for hundreds of American cities covering their residents from the late nineteenth century and throughout much of the twentieth century. They offer a rich source of information about individuals' employment, occupation, and residential choices for a period in which this not available through other sources. Transcribed and linked at scale, following the process outlined in this paper, they would extend the study of careers, firms, and job loss across the whole of urban America in the decades before administrative data is available.

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A Data appendix

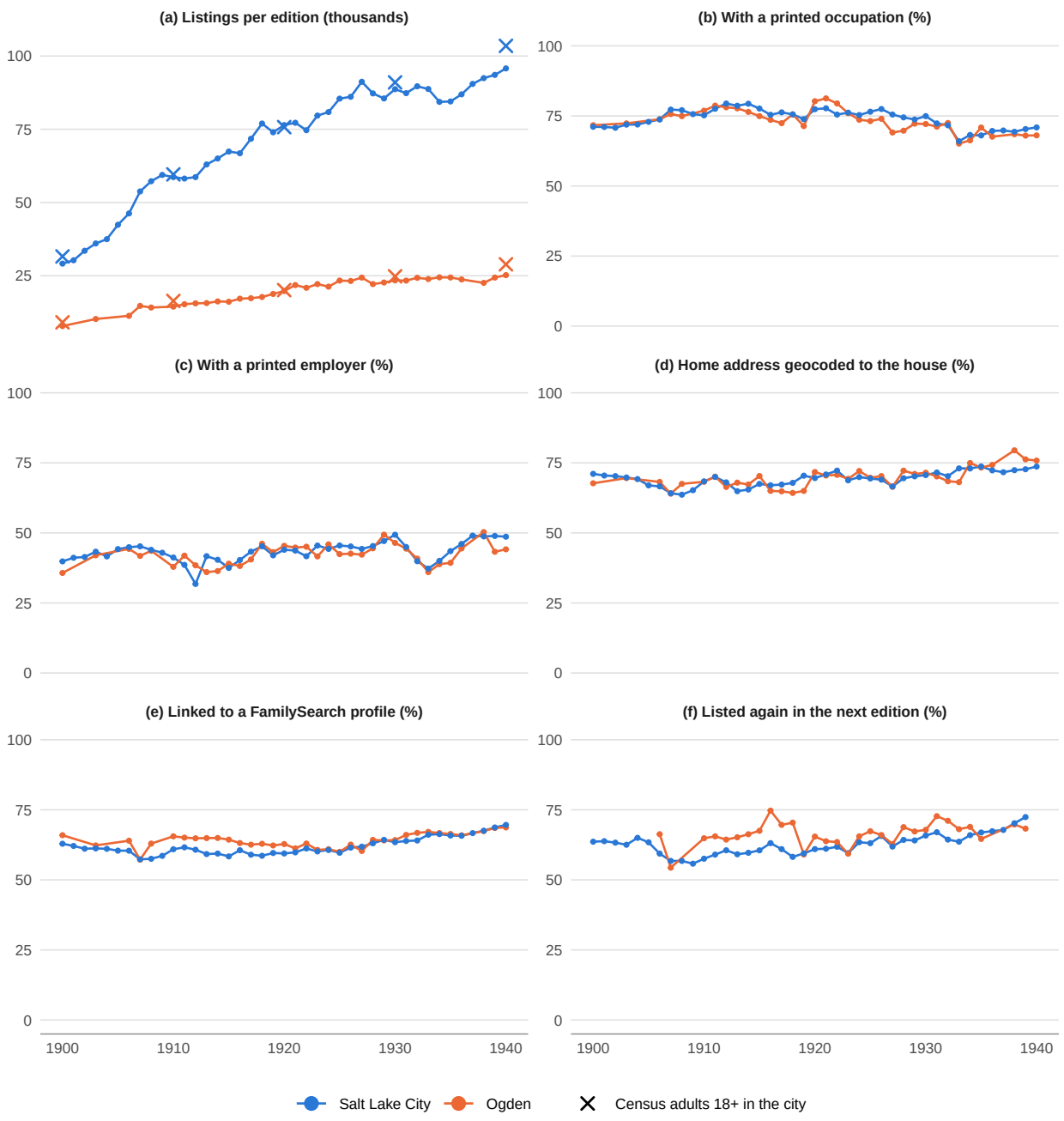


Figure A1: The Salt Lake City and Ogden directories edition by edition, 1900–1940

Notes: One point per edition; lines join consecutive editions, across Ogden's missing editions of 1901, 1902, 1904, 1905, 1909 and 1937. Listings are one row per person per edition, the whole book including its suburban and outlying-town listings, and every share in panels (b)–(f) is a share of them. Crosses in panel (a) are the census population aged 18 and over of the city (IPUMS CITY label). A house-level geocode is an address point or a position interpolated within the block's address range. In panel (f) a listing's person is the FamilySearch pid where linked, otherwise the within-city directory chain; Ogden editions without a successor (1900, 1903, 1908, 1936) and 1940 have no value. Panels (b)–(e) are the per-edition values behind Table 1, panel (f) those behind the first column of Table A3.

A.1 Occupations

Table A1: The eleven IPUMS major occupation groups and the five most common occupations in each

IPUMS major group	Share of sample (%)	Most common OCC1950 titles, in order	Share of group (%)
Professional, technical and kindred	6.8	Teachers (n.e.c.)	15.8
		Lawyers and judges	11.7
		Physicians and surgeons	9.0
		Accountants and auditors	7.3
		Dentists	6.5
Managers, officials, and proprietors	12.3	Managers, officials, and proprietors (nec)	83.7
		Conductors, railroad	8.6
		Officials and administratators (nec), public administration	2.6
		Purchasing agents and buyers (nec)	1.6
		Credit men	0.9
Clerical and kindred	13.7	Clerical and kindred workers (n.e.c.)	61.0
		Bookkeepers	12.1
		Stenographers, typists, and secretaries	3.9
		Mail carriers	3.9
		Agents (nec)	3.3
Sales workers	8.1	Salesmen and sales clerks (nec)	86.9
		Insurance agents and brokers	7.2
		Real estate agents and brokers	2.7
		Hucksters and peddlers	1.8
		Advertising agents and salesmen	1.1
Craftsmen, foremen, and kindred	23.0	Carpenters	14.4
		Foremen (nec)	8.9
		Mechanics and repairmen (nec)	8.0
		Machinists	7.4
		Painters, construction and maintenance	6.5
Operatives and kindred	11.4	Operative and kindred workers (nec)	18.5
		Truck and tractor drivers	16.6
		Mine operatives and laborers	12.7
		Brakemen, railroad	9.9
		Taxicab drivers and chauffeurs	6.5
Service workers, except private household	5.2	Janitors and sextons	21.4
		Barbers, beauticians, and manicurists	14.5
		Guards, watchmen, and doorkeepers	11.1
		Cooks, except private household	9.8
		Waiters and waitresses	7.4
Service workers, private household	0.2	Private household workers (nec)	86.5
		Laundresses, private household	12.8
		Housekeepers, private household	0.7
Laborers, except farm and mine	14.0	Laborers (nec)	81.0
		Teamsters	16.8
		Gardeners, except farm and groundskeepers	2.1
		Garage laborers and car washers and greasers	0.1
		Lumbermen, raftsmen, and woodchoppers	0.0
Farmers	5.2	Farmers (owners and tenants)	99.9
		Farm managers	0.1
Farm laborers	0.1	Farm laborers, wage workers	85.5
		Farm service laborers, self-employed	14.3
		Farm foremen	0.2

Notes: Analysis-sample person-years (FamilySearch-linked men 18–64 with a labor-force OCC1950 code, Salt Lake City and Ogden, 1900–1940). Titles are the IPUMS OCC1950 titles; share of sample is the group's share of all person-years, share of group the title's share of the group's person-years. Groups with fewer than five titles list all of them.

Table A2: The forty most common directory occupation strings, their OCC1950 code, and their agreement with the census

Printed	Normalized	OCC1950 title assigned	Person-years	Census pairs	Same code (%)	Most common census title when different
clk	clerk	Clerical and kindred workers (n.e.c.)*	93,407	9,551	40	Salesmen and sales clerks (nec) (15%)
lab	laborer	Laborers (nec)	84,831	8,083	50	Farm laborers, wage workers (6%)
slsmn	salesman	Salesmen and sales clerks (nec)*	57,908	6,541	71	Managers, officials, and proprietors (nec) (6%)
farmer	farmer	Farmers (owners and tenants)	46,854	4,431	70	Farm laborers, wage workers (12%)
carp	carpenter	Carpenters	33,979	3,794	78	Laborers (nec) (4%)
driver	driver	Truck and tractor drivers*	27,664	2,949	43	Deliverymen and routemen (11%)
hlpr	helper	Laborers (nec)*	25,873	2,682	24	Operative and kindred workers (nec) (18%)
eng	engineer	Stationary engineers*	22,648	2,394	57	Civil-Engineers (6%)
mgr	manager	Managers, officials, and proprietors (nec)	21,983	2,199	61	Salesmen and sales clerks (nec) (7%)
formn	foreman	Foremen (nec)	19,250	1,902	43	Managers, officials, and proprietors (nec) (6%)
mach	machinist	Machinists	18,183	1,956	66	Automobile-mechanics and repairmen (4%)
bkpr	book keeper	Bookkeepers	15,297	1,599	67	Clerical and kindred workers (n.e.c.) (7%)
painter	painter	Painters, construction and maintenance	14,818	1,619	69	Painters, except construction or maintenance (6%)
mech	mechanic	Mechanics and repairmen (nec)*	13,161	1,138	30	Automobile-mechanics and repairmen (22%)
cond	conductor	Conductors, railroad*	12,692	1,266	63	Conductors, bus and street railway (11%)
tmstr	teamster	Teamsters	12,404	1,068	43	Truck and tractor drivers (14%)
firmn	fireman	Locomotive firemen*	11,757	1,025	46	Locomotive firemen (8%)
brkmn	brakeman	Brakemen, railroad	10,631	1,158	72	Conductors, railroad (7%)
chfr	chauffeur	Taxicab drivers and chauffeurs*	9,526	1,089	20	Truck and tractor drivers (30%)
jan	janitor	Janitors and sextons	9,401	904	63	Laborers (nec) (7%)
opr	operator	Operative and kindred workers (nec)*	9,151	968	44	Motormen, street, subway, and elevated railway (6%)
electn	electrician	Electricians	9,073	931	72	Laborers (nec) (4%)
tchr	teacher	Teachers (n.e.c.)	8,399	970	80	Musicians and music teachers (4%)
barber	barber	Barbers, beauticians, and manicurists	7,820	763	86	Laborers (nec) (2%)
miner	miner	Mine operatives and laborers	7,463	844	62	Laborers (nec) (8%)
agt	agent	Insurance agents and brokers*	7,423	703	48	Salesmen and sales clerks (nec) (12%)
repr	repairer	Mechanics and repairmen (nec)*	7,316	606	12	Railroad and car shop-mechanics and repairmen (13%)
contr	contractor	Managers, officials, and proprietors (nec)	7,185	734	59	Carpenters (17%)
swchmn	switchman	Switchmen, railroad	7,106	809	77	Managers, officials, and proprietors (nec) (2%)
pres	president	Managers, officials, and proprietors (nec)	6,851	607	56	Farmers (owners and tenants) (6%)
appr	apprentice	Apprentices, trade not specified	6,212	566	2	Machinists (13%)
plmbr	plumber	Plumbers and pipe fitters	5,803	598	79	Managers, officials, and proprietors (nec) (4%)
mining	mining	Mine operatives and laborers	5,714	487	41	Managers, officials, and proprietors (nec) (29%)
lawyer	lawyer	Lawyers and judges	5,713	612	92	Farmers (owners and tenants) (1%)
insp	inspector	Inspectors (nec)	5,489	471	44	Inspectors, public administration (10%)
blksmith	blacksmith	Blacksmiths	5,239	538	78	Laborers (nec) (4%)
carrier	carrier	Mail carriers	5,145	548	88	Clerical and kindred workers (n.e.c.) (2%)
steno	stenographer	Stenographers, typists, and secretaries	4,992	483	58	Clerical and kindred workers (n.e.c.) (16%)
sec	secretary	Managers, officials, and proprietors (nec)	4,760	462	24	Stenographers, typists, and secretaries (22%)
cook	cook	Cooks, except private household	4,680	476	61	Managers, officials, and proprietors (nec) (4%)

Notes: Analysis-sample person-years, Salt Lake City and Ogden 1900–1940. Normalized = the database's normalized occupation string, the unit to which the census calibration assigns codes; printed = the most common abbreviation behind it. Title = the OCC1950 title assigned; an asterisk marks strings that receive more than one code, because the calibration codes a string separately by employer, employer industry and census era, and the modal title is shown. Census pairs = analysis-sample men with a labor-force occupation in both the directory and the census of the same year; same code = share with the identical three-digit OCC1950 code; the last column is the most common census title among the disagreeing pairs, with its share of all pairs; it can repeat the modal title when the string's other codes are the ones that disagree (fireman: the census records as locomotive firemen men whose listing is coded stationary fireman).

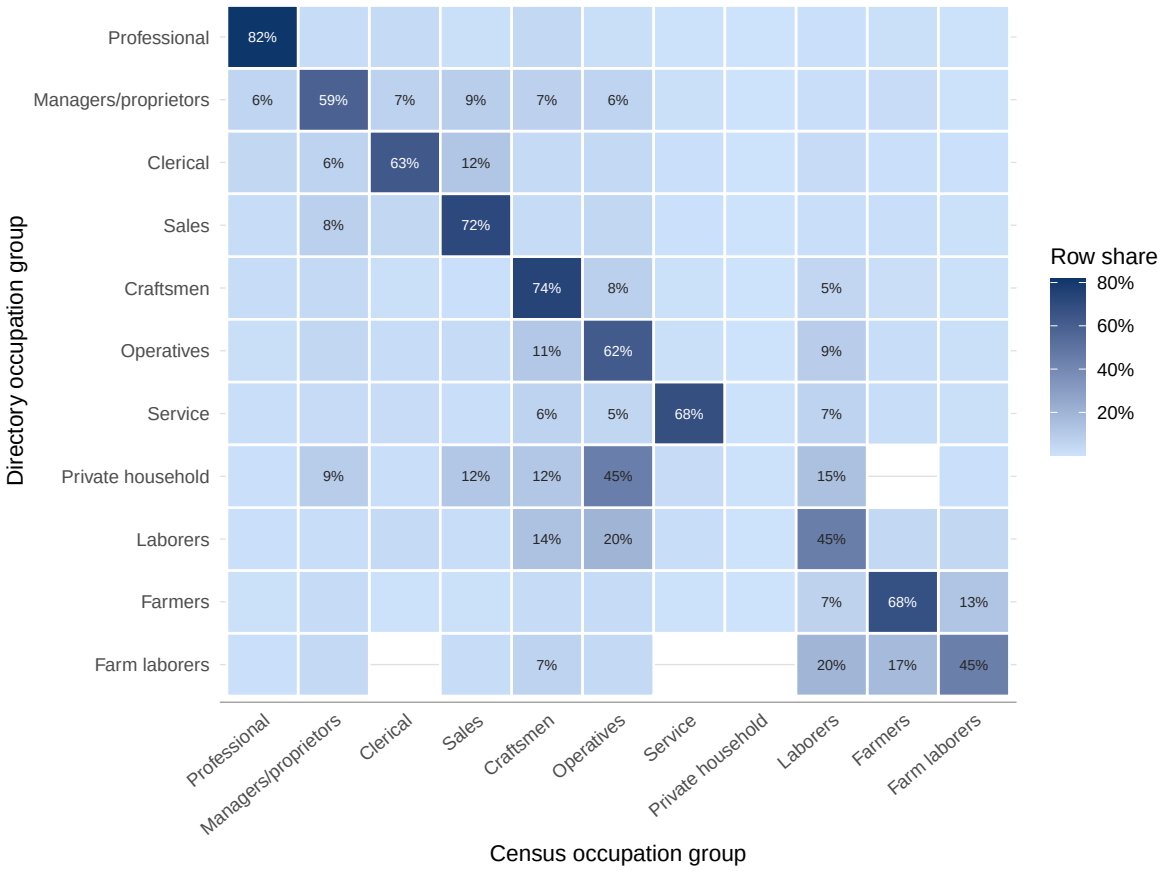


Figure A2: Directory versus census occupation group, same man, same census year

Notes: Analysis-sample men with a labor-force occupation in both the directory and the census of the same year, 1900–1940 pooled, Salt Lake City and Ogden. Rows are the directory's major group and sum to 100 percent; columns are the census's; shares below 5 percent are not printed. The private-household row rests on 176 pairs and the farm-laborer row on 76.

A.2 Addresses

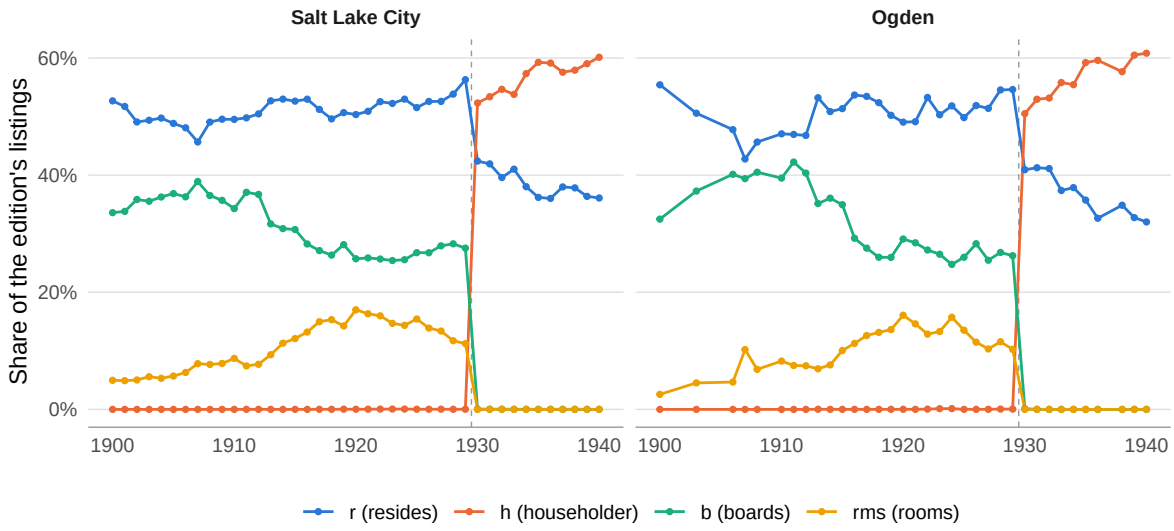


Figure A3: Residence markers printed before the home address, by edition

Notes: Shares of all listings of the edition. Before 1930 the books print 'r' (resides), 'b' (boards) or 'rms' (rooms); from the 1930 editions they print 'h' for the householder and 'r' for every other resident of the household, and 'b' and 'rms' disappear. Listings without an address or without a marker make up the remainder. Markers are read from the address-reference field of the database's listing table.

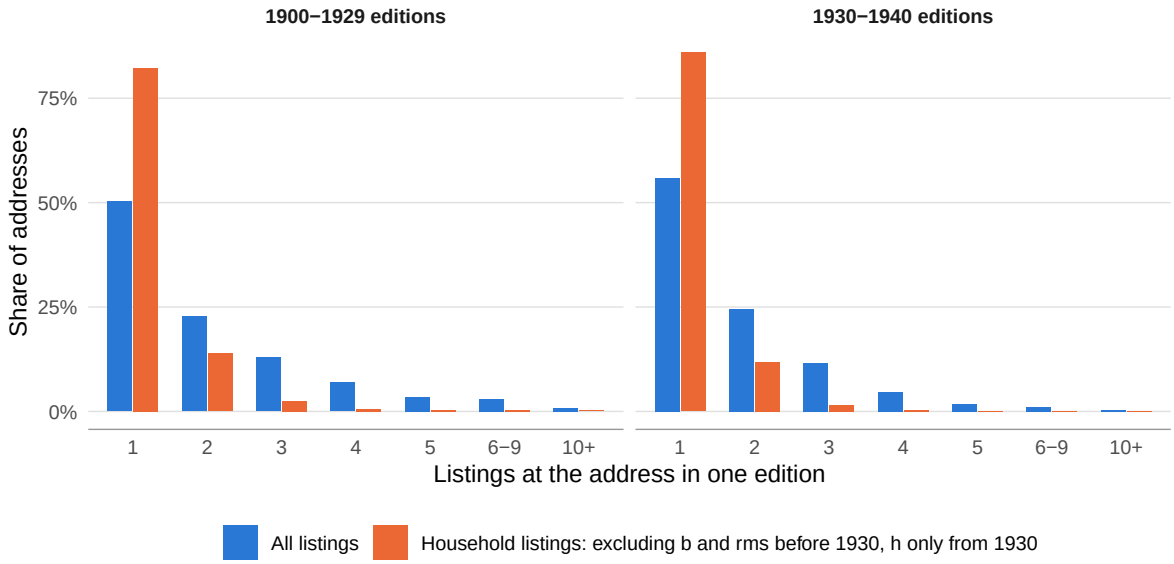


Figure A4: Listings per address in one edition, Salt Lake City and Ogden 1900–1940

Notes: An address is the printed address string within a city and edition, so an apartment house is one address. Household listings drop the listings marked 'b' (boards) or 'rms' (rooms) before 1930 and keep only the listings marked 'h' (householder) from 1930, when the books stop distinguishing boarders and roomers (Figure A3); the series counts the addresses with at least one remaining listing. Mean listings per address: 2.3 (all) and 1.4 (household) before 1930, 2.0 and 1.3 from 1930.

A.3 Linking and coverage

Table A3: Share of listings whose person is found again: in the next edition, in the same book ten years later, and in the next census

	Salt Lake City				Ogden			
	Directory		Next census		Directory		Next census	
	Next ed.	10 years on	Same city	Anywhere	Next ed.	10 years on	Same city	Anywhere
All listings	62.6	29.0	–	–	66.3	30.5	–	–
FamilySearch-linked listings	78.4	45.0	41.9	73.7	80.2	44.6	41.7	75.5
Linked men 18–64	81.8	54.4	43.8	77.0	83.3	54.0	44.3	78.7
Linked men 18–64 with an occupation (analysis sample)	83.7	55.7	45.2	77.5	85.2	55.2	45.0	78.9

Notes: Shares in percent, averaged without weights over base editions. A listing's person is the FamilySearch pid where linked, otherwise the within-city directory chain. Next edition: base editions 1900–1939 whose city has an edition the following year (Ogden lacks 1901, 1902, 1904, 1905, 1909 and 1937, so its editions of 1900, 1903, 1908 and 1936 are excluded); the person has a listing in the city's next edition. The three other columns use the census-year editions of 1900, 1910, 1920 and 1930 as bases and follow the same listings ten years on: listed in the same city's book of the next census year; enumerated in the same city (IPUMS CITY label) in the next census; enumerated anywhere in the complete-count census. Census columns require a FamilySearch link and are undefined for unlinked listings; a listing is enumerated when its pid carries an unambiguous record in the deduplicated linked census file. Men 18–64 are aged by FamilySearch birth year in the base year; the analysis sample adds a labor-force OCC1950 code. The denominator of every row is all listings of the book in the base edition, suburban and outlying-town listings included, so a suburban listing counts as 'same city' in the directory but not in the census. Listings without a FamilySearch link, followed only by their directory chain, are found in the next edition 37.3 percent of the time in Salt Lake City and 42.0 percent in Ogden; a chain breaks when a person moves house and changes occupation or employer in the same year.

Table A4: Directory coverage of the census population, Salt Lake City and Ogden, by census year

		All adults			Men 18–64		
City	Census	Population	FamilySearch (%)	Directory (%)	Population	FamilySearch (%)	Directory (%)
Salt Lake City	1900	31,587	75.5	34.3	13,966	74.1	53.5
Salt Lake City	1910	59,598	72.0	33.4	29,678	66.4	47.5
Salt Lake City	1920	75,733	77.7	33.6	35,386	75.3	50.9
Salt Lake City	1930	90,997	78.6	35.5	41,006	77.7	54.0
Salt Lake City	1940	103,465	82.3	40.6	45,148	82.8	62.4
Ogden	1900	9,094	77.2	33.9	4,308	74.2	52.5
Ogden	1910	16,402	71.2	31.9	8,580	63.3	44.7
Ogden	1920	20,099	82.2	35.6	9,471	80.5	56.2
Ogden	1930	24,796	82.4	37.3	11,269	82.0	58.6
Ogden	1940	28,896	85.1	41.2	12,654	85.3	64.7

Notes: Census residents of the city (IPUMS CITY label) aged 18 and over, one row per person; the directory edition is the census year's. FamilySearch = share with a FamilySearch pid; directory = share of all residents whose pid appears in the city's directory of the edition year. A resident can be matched to the directory only through a pid, so the directory rate is bounded above by the FamilySearch rate; dividing the two gives coverage among the linkable population.

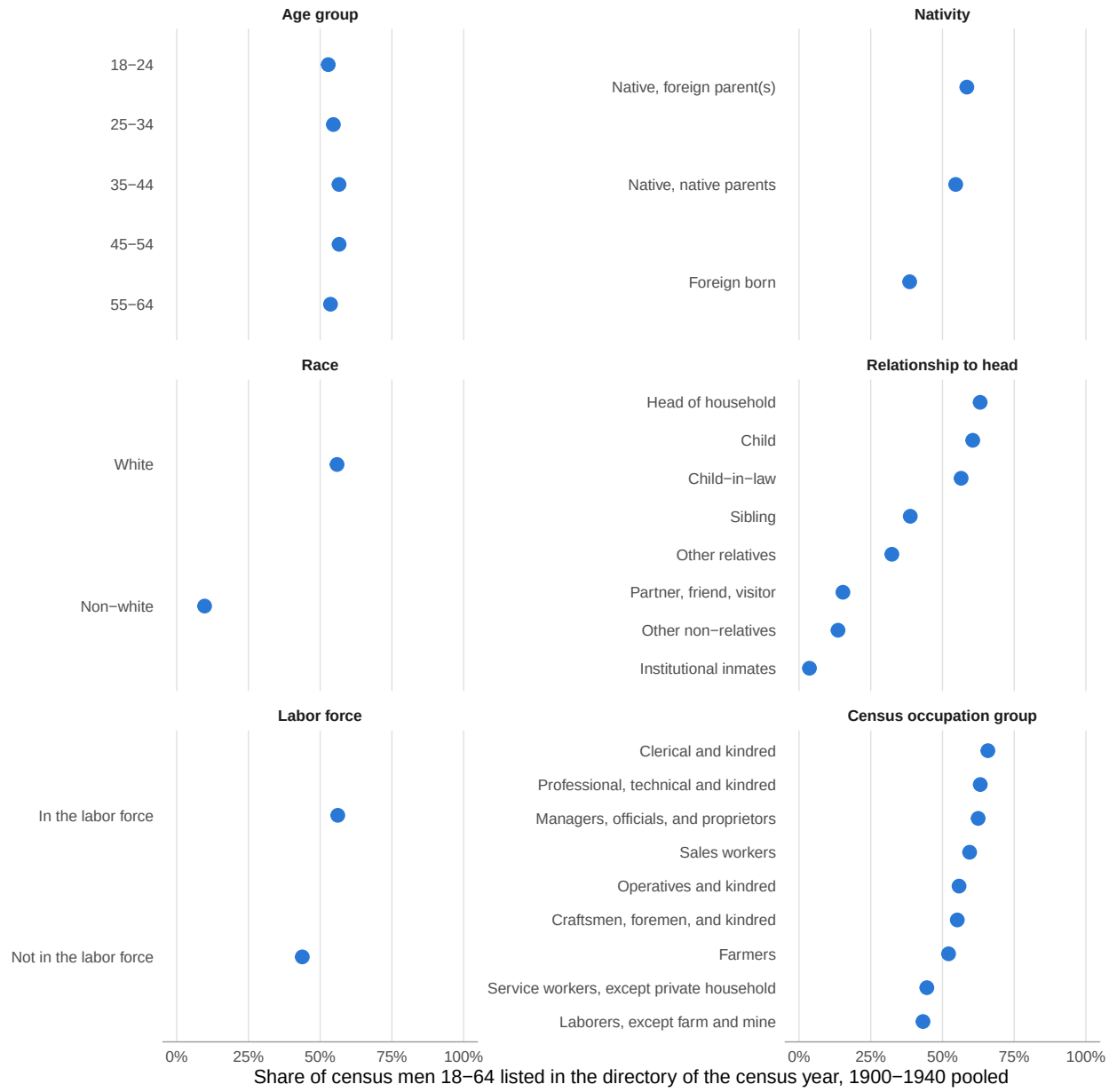


Figure A5: Who is undercovered: share of census men aged 18–64 listed in the directory of the census year

Notes: Census residents of Salt Lake City and Ogden, men aged 18–64, census years 1900–1940 pooled. A man counts as listed when his FamilySearch pid appears in the city's directory of the census year, so the rate folds in FamilySearch linkability. Characteristics are the census's. Cells with fewer than 200 men are omitted, as are the private-household and farm-laborer occupation groups.

A.4 Employers

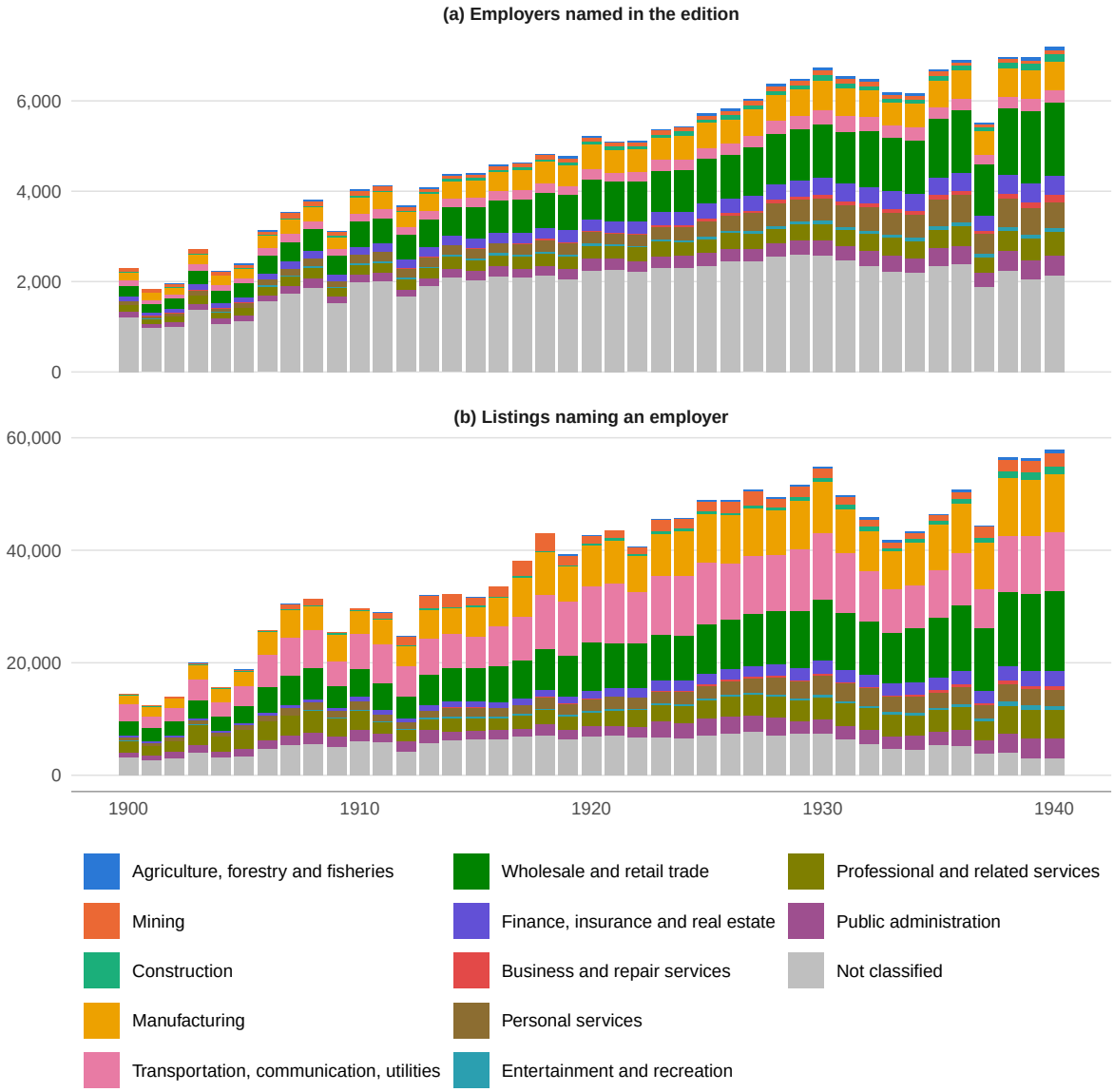


Figure A6: Employers named in the Salt Lake City and Ogden directories, by year and industry

Notes: Employers are the harmonized employers with at least one listing naming them in the edition; panel (b) counts the listings. Industries are the major groups of the 1950 Census industry classification: an employer's industry is the most common 1940 census industry of its linked workers where two or more are observed, otherwise it is assigned from its heading in the classified business directory or from its name. Unclassified employers, mostly small firms named after a person, are 41 percent of employer-years but 14 percent of the listings that name an employer.



Figure A7: What the directory's employer field means: directory state against census status, 1940

Notes: Men aged 18–64 in Salt Lake City or Ogden (IPUMS city label) in the 1940 census, linked to FamilySearch and found in the 1940 Salt Lake City or Ogden directory, by their state in the directory (rows; number of men in parentheses) and their status in the census (columns); each row sums to 100 percent. Listed with an employer: an employer is printed. Likely self-employed: no employer printed, and either named as the owner, proprietor or partner of a firm in the same edition (81 percent of the row) or listed with an occupation string that the census records as self-employed at least half the time. Census status from the employment status and class of worker of the full-count census. Employed, other: unpaid family workers, the armed forces and employed men whose class of worker is illegible.

B The role of firms: additional results

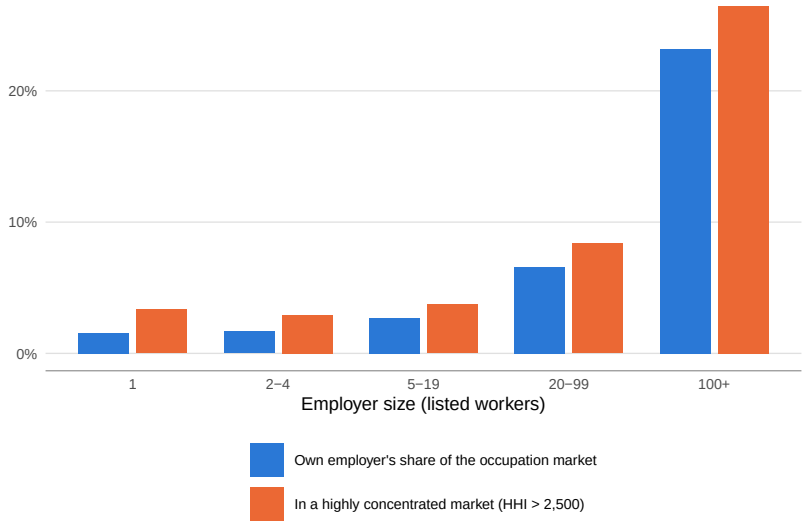


Figure A8: Concentration of occupation markets by employer size

Notes: Listings with a labor-force occupation that name an employer, Salt Lake City and Ogden, 1900–1940 pooled (the 1912 editions left out), by the size of the named employer in the city and edition (1: an employer named by a single such listing). Own employer's share: the named employer's share of the listings in the worker's occupation market (a city, a year and an OCC1950 code), averaged over listings. Highly concentrated: an employer Herfindahl–Hirschman index above 2,500. Of the listings at employers of 100 or more, 89.4 percent were in markets with at least five other employers. Workers whose employer is not printed and the self-employed are not counted as part of any employer's market.

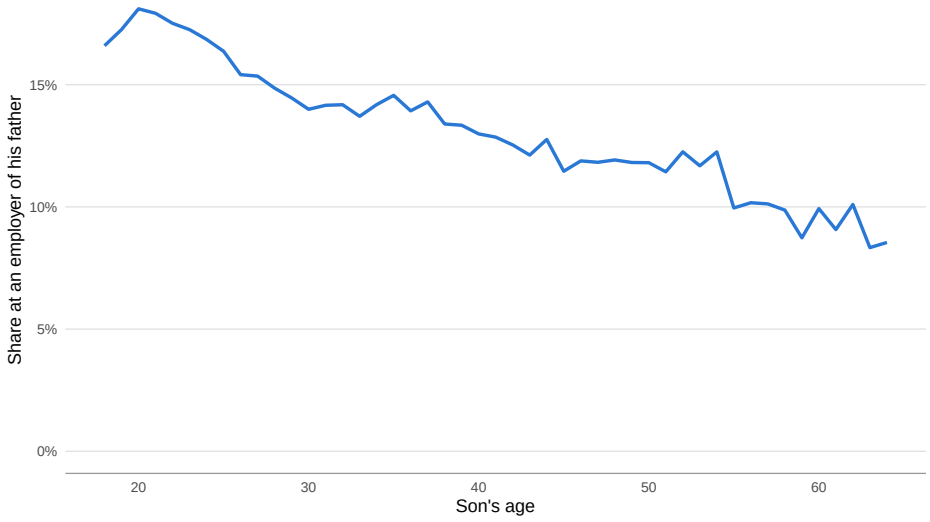


Figure A9: Sons at an employer of their father, by age

Notes: Analysis-sample person-years with an employer of men whose FamilySearch father had been listed with an employer in any of the four cities' directories by year t (244,981 person-years), 1900–1940 pooled; ages with at least 100 person-years. A son is at an employer of his father if his employer in t is one his father had been listed with in t or earlier. If each son is assigned the father of a randomly drawn son of the same year, the share is 1.4 percent.

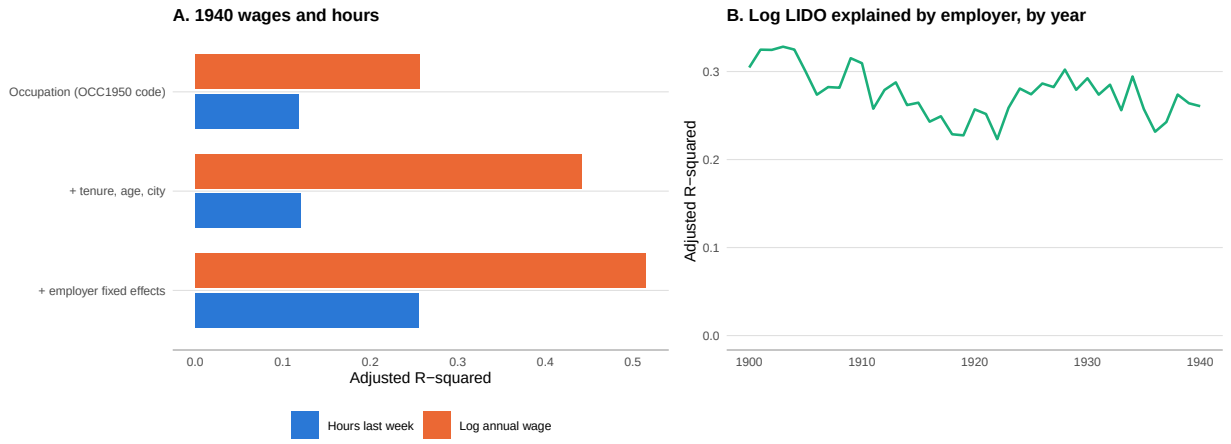


Figure A10: How much do employers explain?

Notes: Panel A: adjusted R^2 of nested regressions of 1940 log annual wage and hours worked in the census week, for the men of Table 2 at employers with two or more such men (hours: men at work). Panel B: adjusted R^2 of employer fixed effects in a regression of log LIDO, analysis-sample men listed with an employer that has two or more such men in the year.

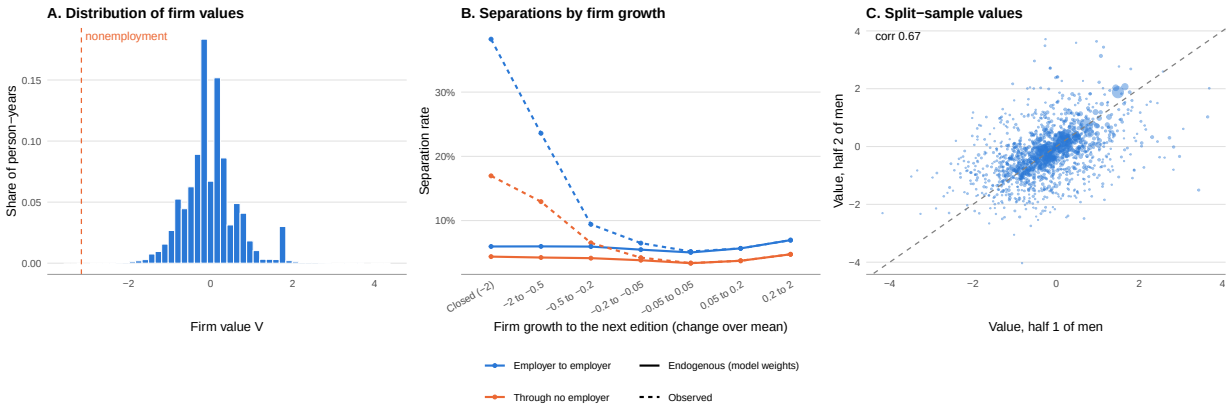


Figure A11: Firm values from revealed preference: diagnostics

Notes: The estimates of Figure 5. Panel A: distribution of firm values over person-years; the dashed line is the value of having no employer, 3.15 below the mean firm. Panel B: annual separation rates of sample men by the firm's growth to the next edition (change over the mean of the two counts), observed and as weighted by the model's endogenous share; at closing firms 38.2 percent of men move directly to another employer and 17.0 percent through a year without one, of which the model treats 6.0 and 4.4 points as endogenous. Panel C: values estimated separately from two random halves of the men (1,298 firms in both sets; correlation 0.67, rank correlation 0.43), points sized by person-years.

Table A5: The ten largest employers, 1910–1940

	1910	1920	1930	1940
1	Oregon Short Line R R Co (SLC), 1,167	Southern Pacific Lines (Ogden), 1,769	Oregon Short Line R R Co (SLC), 1,752	Utah Copper Co (SLC), 2,147
2	Denver & Rio Grande Western R R Co (SLC), 1,072	Denver & Rio Grande Western R R Co (SLC), 1,741	Denver & Rio Grande Western R R Co (SLC), 1,530	Union Pacific R R Co (SLC), 1,518
3	Utah Light & Ry Co (SLC), 895	Oregon Short Line R R Co (SLC), 1,713	Southern Pacific Lines (Ogden), 1,308	Denver & Rio Grande Western R R Co (SLC), 1,454
4	Southern Pacific Lines (Ogden), 709	Utah Copper Co (SLC), 1,114	Utah Copper Co (SLC), 1,162	American Smelting & Refining Co (SLC), 1,415
5	Fort Douglas (SLC), 680	American Smelting & Refining Co (SLC), 953	American Smelting & Refining Co (SLC), 992	Ogden Union Railway & Depot Co The (Ogden), 906
6	Z C M I (SLC), 346	Ogden Union Railway & Depot Co The (Ogden), 571	Ogden Union Railway & Depot Co The (Ogden), 982	Southern Pacific Lines (Ogden), 771
7	Union Pacific Railroad Co (Ogden), 262	Mountain States Telephone & Telegraph Co (SLC), 546	Mountain States Telephone & Telegraph Co (SLC), 764	Utah Power & Light Co (SLC), 698
8	Mountain States Telephone & Telegraph Co (SLC), 217	Z C M I (SLC), 530	Z C M I (SLC), 731	Z C M I (SLC), 695
9	Keith-O'Brien Inc (SLC), 197	Utah Light & Traction Co (SLC), 446	Mountain Fuel Supply Company (SLC), 571	United States Smelting Refining & Mining Co (SLC), 669
10	Postoffice (SLC), 184	Keith-O'Brien Inc (SLC), 416	Union Pacific Railroad Co (Ogden), 486	Mountain States Telephone & Telegraph Co (SLC), 629

Notes: Harmonized employers ranked by the number of listings with a labor-force occupation that name them in the city's directory of the year; an employer in Salt Lake City and in Ogden counts as two employers. Names as printed in the directory.

C Firm closures

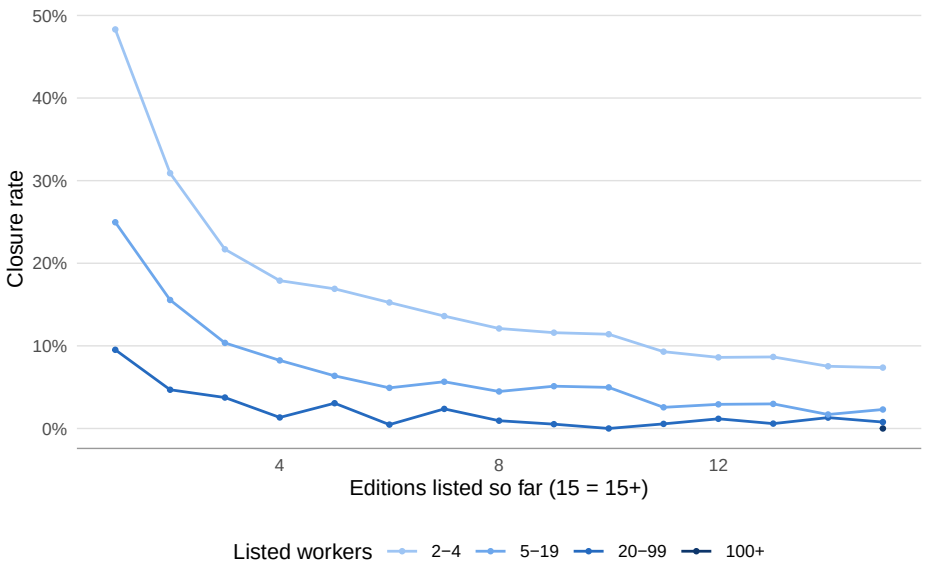


Figure A12: Closure rate by firm age and size

Notes: Closure rate (Figure 6) by the number of editions in which the firm has been listed and by its listed workers in the year, for firms first listed from 1901, so that their age is observed; government and public bodies left out.

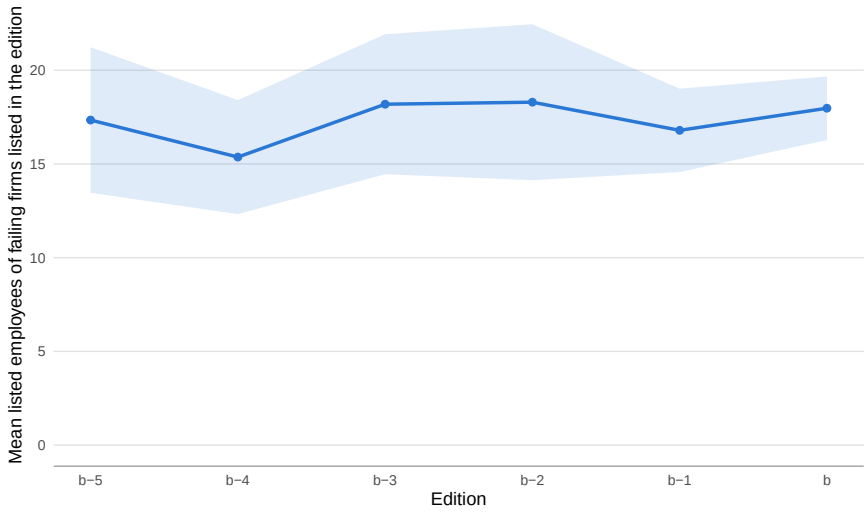


Figure A13: Listed employees of failing firms in the editions before closure

Notes: The 508 employers named by ten or more listings in a base edition b of the Salt Lake City or Ogden directory, 1900–1937, that close after b (absent from the next three editions; government and public bodies, absorbed employers and employers with a successor left out). Listed employees: listings naming the employer in the edition. Mean over all failing firms, an employer not yet listed counting zero (63 percent were listed in $b-1$ and 31 percent in $b-5$), and over the failing firms listed in that edition, with 95 percent confidence intervals; editions are counted within the city, so Ogden's missing early books are not treated as absences.

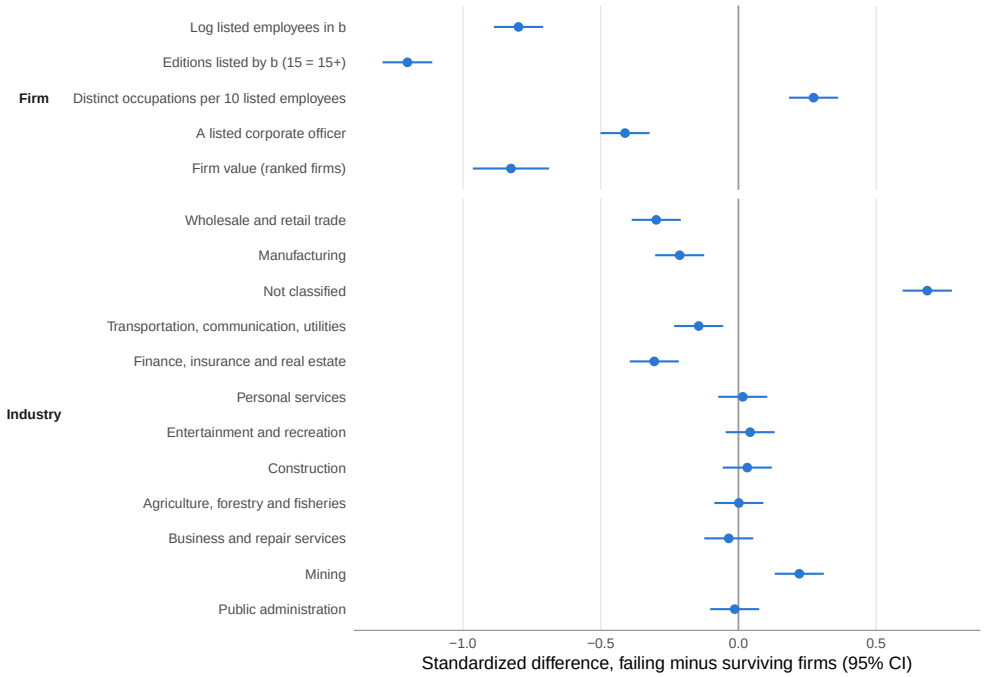


Figure A14: Differences between failing and surviving firms in the year the failing firms closed

Notes: Standardized differences in means, with 95 percent confidence intervals, between the 508 employers that close after a base edition b (Figure A13) and the 10,973 employer-editions with ten or more listed employees that survive (at least 75 percent of the listed employees kept over the next three editions and still listed at $b + 8$); employers that shrink by more than a quarter are in neither group. Listed employees: listings naming the employer in b. Firm value: the revealed-preference value of Section 3, estimated without the moves out of closing firm-years, for the firms in the ranked set. Industries are the major groups of the 1950 Census classification.

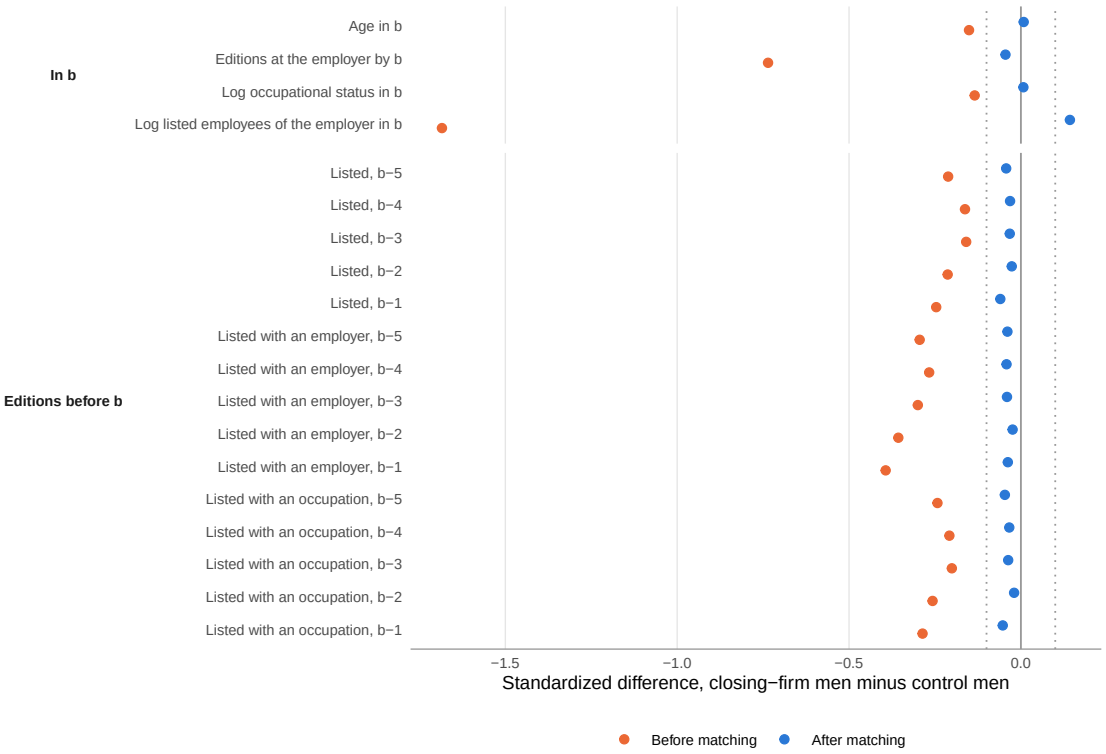


Figure A15: Men at closing and surviving firms before and after matching

Notes: Standardized differences in means between the men listed with an occupation at a closing employer in the base edition b and the men at surviving employers of the same stacks (Section 4.2): before matching, 3,297 closing-firm men against 292,726 comparison man-editions; after matching, the 3,038 matched closing-firm men against their 4,934 matched comparison man-editions, weighted by the number of closing-firm men each serves. Dotted lines at ± 0.1.

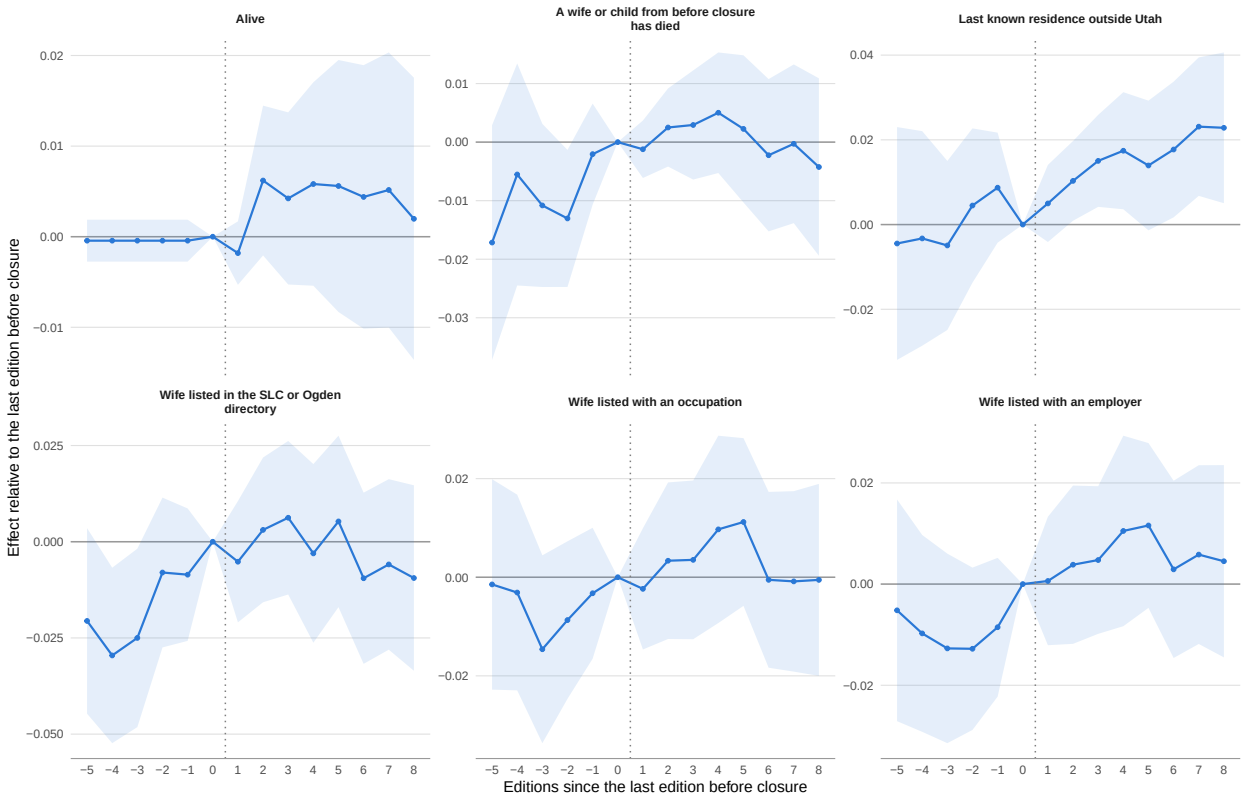


Figure A16: Effects of firm closure on survival, deaths in the family, residence outside Utah and the wife's directory listings

Notes: The design and sample of Figure 7, in calendar years $b-5$ to $b+8$. Alive: no recorded death by the year, all person-years kept; men without a recorded death (3 percent) count as alive. A wife or child has died: among the men with a wife (a dated marriage by b) or a child born by b (5,868 man-stacks), whether any of them has died by the year; this is the one outcome whose paths before b differ (joint test $p = 0.001$: the closing-firm men's relatives had died less often in the years before b). Residence outside Utah: the last known residence of Figure 7 with the state as the boundary. Wife: the man's wife in b, from the earliest dated marriage still intact in b (5,406 man-stacks), listed in the Salt Lake City or Ogden directory, with an occupation and with an employer, in the editions of the man's base city to 1940 as in Figure 7; years after the wife's death are dropped and years after the husband's kept.

Table A6: Balance of exit and survivor men before and after matching on 1920–1928 career paths

Variable	Exit	Survivors, all	Survivors, matched	Std. diff. before	Std. diff. after
Age in 1929	34.397	36.200	34.001	-0.158	0.054
Editions at the 1929 employer	2.552	3.944	2.920	-0.513	-0.108
Log status 1929	3.248	3.268	3.258	-0.093	0.047
Log listed employees of the 1929 employer	3.428	4.986	3.209	-1.259	0.135
Listed, 1920	0.450	0.484	0.458	-0.068	0.002
Listed, 1924	0.615	0.657	0.685	-0.088	-0.146
Listed, 1927	0.792	0.823	0.824	-0.080	-0.105
Listed with an employer, 1920	0.284	0.328	0.301	-0.097	-0.018
Listed with an employer, 1922	0.308	0.356	0.362	-0.104	-0.087
Listed with an employer, 1924	0.384	0.480	0.501	-0.195	-0.223
Listed with an employer, 1926	0.521	0.590	0.537	-0.140	-0.019
Listed with an employer, 1927	0.539	0.651	0.566	-0.231	-0.046
Listed with an employer, 1928	0.657	0.709	0.636	-0.111	0.059
Listed with an occupation, 1920	0.362	0.416	0.359	-0.110	0.034
Listed with an occupation, 1924	0.467	0.544	0.573	-0.154	-0.198
Listed with an occupation, 1928	0.706	0.743	0.720	-0.082	-0.019

Notes: Propensity score: logit of exit on age (quadratic), tenure, 1929 status, log listed employees of the 1929 employer, city, 1929 IPUMS group, 1929 employer industry and indicators for being listed, listed with an employer and listed with an occupation in each edition 1920–1928. Each exit man is matched to up to three survivor men in the same city and 1929 group within 0.2 standard deviations of the logit score, with replacement; matched survivors are weighted by the number of exit men they serve. Rows for a year are shares of men in that edition.